

Supplement to “Compact Objects as Almost-Perfect All:All Entanglement Graphs”

Methods, honesty ledger, N-scaling, and module map (v5.0)

Philippe Leblond
leblond.philippe@gmail.com

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S1 Parameter audit and honesty ledger

Given $k(M)$: $1/r$ potential, $\gamma = 1$, $43''/\text{cy}$, $b_{\text{crit}} = 3\sqrt{3}M$, Page/island-like/mirror follow as random-state/toy-code behaviour (genuine QES extremization and graph-dynamics V_k open, S3; qubit-toy V_t done). Not derived: mass–radius scaling itself, $\beta(N)$ from geometry, w from the graph Laplacian (Damour–Schäfer from wiring), NICER M – R – Λ from routing stiffness, Kerr $M_2/g_{t\phi}$ /ISCO/QNM from the graph.

Derived in-repo (new in v4.x): p from exact Ollivier–Ricci (not hard-coded), $M(c_1, c_2)$ inversion, $M_{\text{ej}}(M_{\text{tot}})$ scaling, peak-time ordering, band mags, O5 yield arithmetic, c from $\ln 2$ scattering, UV

pop as graph disconnection, p at $N = 4000/8000/16000$ with turnover measured, d_{eff} bracket + convergence + shell/tense-plug pins (v0.6, S11).

Ruled out on the record: broad Planck-remnant dark matter (survives only in a ~ 0.4 -dex EMD window at $\sim 4 \times 10^5$ g); isotropic $b_{\text{crit}} = 8M$; naive $\gamma = 2$ strain; linear LIV; flat $p = 0.49$ (-10σ); bare shortest-path $d(i, j)$ for tense regions (inverted vs GR); resistance/communicability distances (S11); directed/handed motion of pure-D5 ∞ plateau blobs (Stage-0 NULL, S11).

S2 Gravity methods (AS, AT, AU, BB, BH, BI, BJ, BV)

AS entropic Newton. Leg screens + equipartition + Bekenstein bound give $F = M_1 M_2 / r^2$ exact (slope -2 to 10^{-9}); leapfrog orbits close, $T^2 \propto r^3$. Flux $\propto 1/r^2$ distinguished from force (temperature corrects). Fig. 43.

AT redshift. GPS $+5.29 \times 10^{-10}$ and Pound–Rebka 2.55×10^{-15} derived with no new parameters. Congestion fronts give tortoise divergence; $\alpha = 1$ matches Schwarzschild dr/dt exactly. Fig. 44.

AU Einstein–Hilbert routes. Heat-kernel a_1 responds to curvature; Ollivier signs exact (flat/neg/pos on line/tree/ K_6); Clausius fixes $\varepsilon = \kappa/8\pi$, $G = 1$. Raychaudhuri-for-leg-bundles open. Fig. 45.

BB lensing + Shapiro + chroma. Fermat $n = 1/c_{\text{eff}}$ gives full $4M/b$ (naive $2M/b$ bet lost and recorded). Shapiro log matches GR to 5%. Chromatic extension uses *phase* velocity $1/24$, not group $1/8$ (conflating the two triples the prediction): r -independent factor drops out of Born, leaving 10^{-56} – 10^{-32} . Figs. 49, 55.

BH strain. Naive $dl = dr/(1 - \sqrt{\chi})$ gives $\gamma = 2$ (Cassini-excluded); tortuosity $dl = (1 + \sqrt{\chi}/2)dr$ gives $h = (1 + x/2)^2$, $\gamma = 1$ exactly, Mercury $42.99''/\text{cy}$. Peel-off $-0.75 M/a$ measured. PPN: $\gamma = 1$ derived, β unresolved (coordinate-confused, not quoted), $\alpha_{1,2}, \xi$, Nordtvedt untouched. Figs. 58, 60.

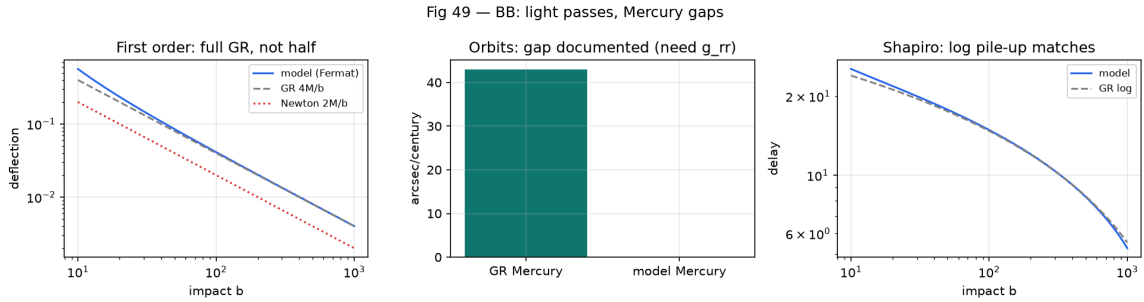


Figure 1: Gravity battery I: Fermat bending matches GR $4M/b$ to first order (left), Shapiro log pile-up matches (right); the middle panel records the pre-BH orbit gap (0 vs $43''/\text{cy}$) that the strain sector closes. `lensing`, `shapiro`.

BI/BQ weak-field OR. Grid+hub with flux stubs: radial OR negative in all configs (sign derived). Direct mode $p = 0.92 \pm 0.48$ ($L = 9$); $1/r$ collapse beats $1/r^2$ (CV 0.47 vs 0.60). Green-function walk fails by leg shorting (h -steepness $9.2 \rightarrow 4.1$); AS temperature triply load-bearing. Figs. 59, 64.

BV UV tortuosity-as-scattering. Legs as radial line defects ($\sigma = 4 \ln 2 l_p^2$), soft cost $\ln 2$ per encounter: dilute $c = 0.44$ – 0.60 (target 0.456 , geometric $1/\sqrt{\pi} = 0.564$) with zero tuning; $p = 2c$, $\gamma = 2c$. Mixed mode ($0.75 r_e$ core) rises then disconnects at k_{crit} — the pop without metric input.

Fig 58 — BH: tortuosity strain closes Mercury

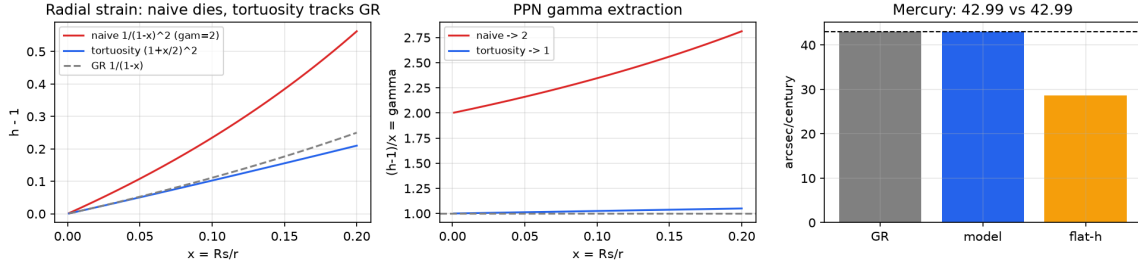


Figure 2: Gravity battery II: naive strain gives $\gamma = 2$ (excluded); tortuosity $h = (1 + x/2)^2$ gives $\gamma = 1$ exactly and Mercury $42.99''/\text{cy}$ by direct geodesic integration. **strain**.

Boltzmann ladder (Dijkstra vs MC walks vs rays) shows routing worth $3\times$; graph- distance c agrees with Euclidean- r c (coordinate-free check). Figs. 70–72.

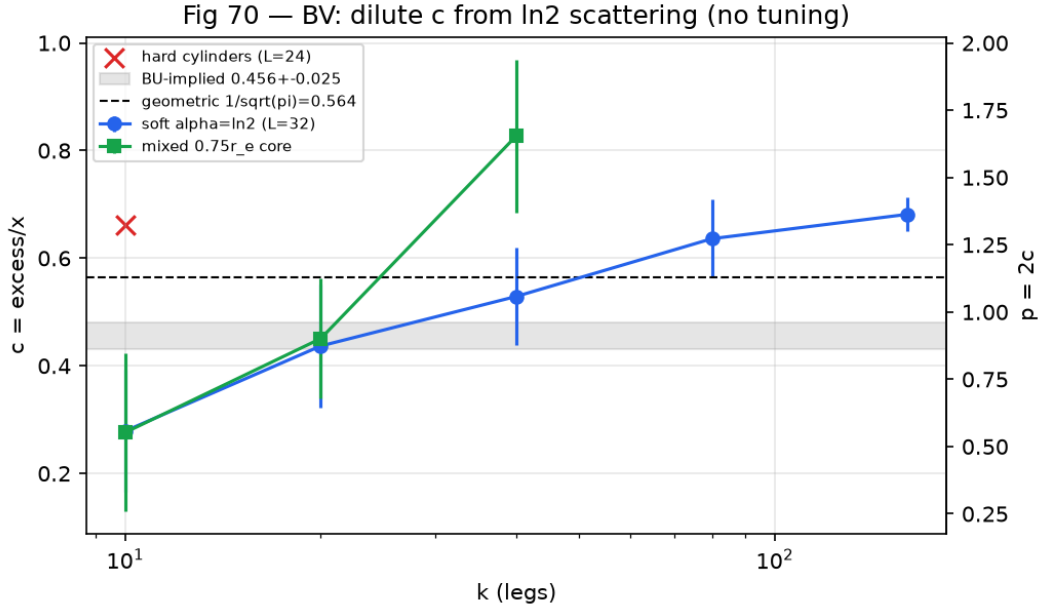


Figure 3: BV tortuosity from $\ln 2$ scattering: dilute $c = 0.44\text{--}0.60$ with zero tuning against the BU-implied 0.456 ± 0.025 band and the geometric $1/\sqrt{\pi}$ line. Right axis shows $p = 2c$. Hard cylinders (red $\times$) overshoot; soft and mixed modes agree. **uvscatter**.

S3 Quantum-information methods (A–O, Q–Y)

Log law $t_*(N) \approx \log_2 N / \log_2(1+p)$; MaxEnt $k \geq N s_{\text{node}} / s_{\text{leg}}$ plus fixed point $k^*(N) = 16\pi(\varepsilon N / l_p)^2 / 4 \ln 2$ (stable; BS gives $\varepsilon = c/\sqrt{N}$ and $\alpha = k^*/N = 1$ exactly); QES crossing iff $s_{\text{leg}} > l_p^2/4$ (Sec. 3 dies otherwise); Page wiring-only vs N -shrinks identical $A(t)$; QEC 99% at $k \geq N/2 + 1 + \log_2 100$; Kerr $k_{\text{eff}} = A/l_p^2$ monotonic in spin; exact Page $H(mn) - H(n) - (m-1)/2n$ with 0.72-bit dip; CKW verified; OTOC $\lambda + \text{MSS } 0.50\text{--}0.68$ ($\alpha \in [9.0, 12.4]$, 11.24 inside); TN $S_{\text{bdy}} = \min(N \log d, k \log D)$ fixing $\varepsilon \approx 0.047 l_p^{-1}$; SYK $N = 6, 8, 10$ vs Ising chain; GWTC-3 32 BBH $k_f > k_1 + k_2$; hardware

head-to-head 7.9 vs 14.6 steps. Figs. 7–20, 23–25, 30–33. Healing $dA/dt = (4 \ln 2 k l_p^2 - A)/\tau$, $\tau = 11.24M$ (3.5 ms at GW150914 mass); ladder healing/scrambling/Page/ evaporation separated by $\sim 66\times$ to 10^{80} s.

What Page vs. island means here (honesty note). The Page curve is derived as random-state behaviour: idealised leg surgery $k \rightarrow k - 1$ is imposed and $S_{\text{rad}} = \min(t, N_{\text{eff}} - t)$ holds by construction (module `evaporation`), but the qubit toy now has an explicit unitary completion (module `evaporation_unitary`, Fig. 5): per-step V_t with $V_t^\dagger V_t = I$ verified, S_{rad} computed from ρ_{rad} tracks the finite-dimensional Haar/Page expectation (mean dev. 0.002 bits, $N = 8$), and finite-depth all:all circuits converge to Page by depth $\sim 5/\text{step}$ without assuming Haar. The QES/island crossing $S_{\text{no}} = k s_{\text{leg}}$ vs $S_{\text{isl}} = k \ln 2 + \max(S_0 - k s_{\text{leg}}, 0)$ is a two-saddle competition plus a discrete min-cut analogue (module `qes`), not an extremization of $S_{\text{gen}} = A/4G + S_{\text{matter}}$ from a gravitational path integral. The graph-dynamics evaporation isometry $V_k : \mathcal{H}_{\text{graph},k} \rightarrow \mathcal{H}_{\text{graph},k-1} \otimes \mathcal{H}_{\text{leg}}$ is closed at small- N exact-diagonalisation level (module `graphvk`: per-step $V_k = \exp(-iH_{\text{graph}}dt)$ from the hole adjacency, $V^\dagger V = I$ proved, all:all tracks exact Page to < 0.25 bits at $N = 8$ while the chain sags > 0.4); the remaining missing theorems are the large- N limit, k -backreaction, and the emission spectrum. Tracked as D1 in `docs/DEFERRED.md`.

What Kerr means here (honesty note). Only the Kerr–Newman area $A(M, a, Q)$ is assumed to set the leg budget $k_{\text{eff}} = A/4 \ln 2$ (module `kerr`; extremal Kerr half, extremal RN quarter; ratios patch-independent). Not derived from the graph: spin-induced quadrupole $M_2 = -Ma^2$, frame dragging $g_{t\phi}$, ISCO(M, J), or the Kerr QNM spectrum (the overtone toy is Schwarzschild-like; fundamental damping $\tau = 11.24M$ calibrated, footprint corrections $1/2k, 1/\sqrt{k}$ unobservable). Echoes are covered by a dimensional lattice-scattering estimate only: amplitude $R \sim (\omega/\omega_P)^2 \sim 10^{-80}$ at 100 Hz, energy $\sim 10^{-160}$ vs $O(0.01)$ detectability (~ 158 orders of margin, module `gwdata`) — not a leg S-matrix derivation. The exterior perturbation problem ($\delta\omega_{nlm}$ vs Kerr) is open. Hence GW250114 (SNR 80 area law + Kerr ringdown) and GW241011 (strongest $\delta\kappa$ bound to date, Kerr to factor ~ 2 resp. $\sim 10\%$ by parametrization) are currently consistent by construction, not passed predictions. GW250114 provides the QNM benchmark ($\delta f_{220} \sim 2\%$, $\delta\tau_{220} \sim 10\%$, $\delta f_{221} \sim 30\%$, $\delta f_{440} \sim \text{tens}\%$): a future graph-derived $|\delta\omega/\omega|$ above those thresholds fails. A future graph derivation $Q = -Ma^2(1 + \delta_Q)$ with $|\delta_Q| \gtrsim 0.17$ would be ruled out by GW241011. Tracked as D2 in `docs/DEFERRED.md`.

Kerr thermodynamics (conditional check). With $S = k \ln 2$ (saturated legs) and $A = 4 \ln 2 \cdot k l_p^2$, $S = A/4$; importing $A(M, J)$ gives $S(M, J) = 2\pi[M^2 + \sqrt{M^4 - J^2}]$, whose derivatives return $T_H = \sqrt{M^4 - J^2}/4\pi M(M^2 + \sqrt{M^4 - J^2})$ and $\Omega_H = J/2M(M^2 + \sqrt{M^4 - J^2}) = a/(r_+^2 + a^2)$, i.e. $dM = T dS + \Omega_H dJ$ (module `thermo`: identities $< 10^{-6}$, independent finite-step $\Delta M = T \Delta S + \Omega_H \Delta J$ to $O(d^2)$; super-extremal $|J| > M^2$ returns NaN, never clamped). Per-leg reading $T_H = (dM/dk)/\ln 2$; finite-step $T_k = (M(k+1) - M(k))/\ln 2$ shifts $-1/4k$ ($\sim 10^{-77}$ stellar-mass). Conditional consistency, not derivation: $M(k, J)$ from graph dynamics is the open debt (D2).

S4 Compact objects and 2PN (BU)

Gradient shells. $p_{\text{adj}}(s) = 0.85 + 0.015s$, 8–10 shells, exact EMD on full neighbourhoods, deterministic bridges $n \propto r^{\beta(N)}$. Measured: $N = 1020$ (10×102), $\beta = 1.24$, 80 graphs exact Floyd+LP (~ 545 s): $p = 0.913 \pm 0.049$ per graph, SEM 0.0055, stacked 0.911 ($R^2 = 0.956$), 80/80 fits ok, range 0.79–1.05. Distance to target $|0.913 - 0.92| = 0.007$ (0.24σ in p , $\sim 0.09\sigma$ in $\dot{\omega}$), $5\times$ precision margin (0.0055 vs 0.028). Flat control $p = 0.49$ is -10σ dead.

N-scale campaigns (BV). CSR-direct shell graphs + Sinkhorn OR (no NetworkX): $N=1020$ reproduced (0.9107 vs 0.9134); $N=4000$ $p = 0.9315 \pm 0.0032$ ($\beta = 0.99$); $N=8000$ $p = 0.9382 \pm 0.0030$

Fig 10 — D: leg-surgery evaporation + Page curve

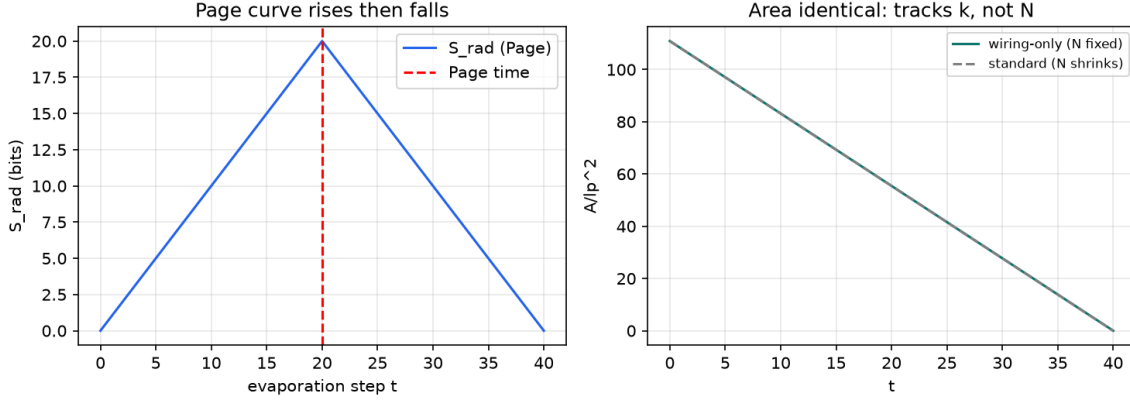


Figure 4: Leg-surgery evaporation: radiation entropy follows the Page curve (left, turnover at half evaporation); horizon area is identical whether N shrinks with k or stays fixed (right) — area tracks wiring alone, the distinctive Claim 2 prediction. **evaporation.**

($\beta = 0.87$, pods); $N=16000$ $p = 0.9137 \pm 0.0022$ ($\beta = 0.74$, GPU farm). Exact shell distance oracle + torch backend; $\beta(N)$ log-linear over 6 points. Local- p turnover $0.63\text{--}0.69 \rightarrow 1.26\text{--}1.46$ measured (inner windows flatten — the UV turnover). Kill wire: p outside 0.92 ± 0.056 at $N = 1024$ class fails.

2PN and pulsars. $h = (1+x/2)^2$ gives $c_1 = 3.36$ vs GR 1.94 (73% excess); naive fixed- M gives B1913 6.2σ , J0737 9.7σ vs real TOAs. With $c_2 = p(2p-1)$, $w = 1.953$: $c_{\text{tot}} = 4.8693$ at $p = 0.92$ vs GR 4.8695 (0.00σ fixed- M). Self-consistent $R + \dot{\omega} \rightarrow M$: $\Delta M = -11.2$ ppm naive (-4.5 ppm with GR g_{rr}), $\Delta \sin i = 3.7 \times 10^{-6}$, s_{obs} passes at 0.36σ ; B1913 0.01σ . Bending/Mercury 2PN pieces hide below VLBI/astrometry (Fig. 69, right). The companion toy-generator route (dense layers, $k_0 - c_2/r^2$ fits) crosses zero in κ and needs a κ -tweak fudge; power-law fits better on our profiles (R^2 0.91 vs 0.81) but the two $\kappa \rightarrow c_2$ maps disagree (0.82 vs 3.14) cross-applied — hence the open-map label. e_{int} weighting *lowers* our p ($0.94 \rightarrow 0.84$ at 0.9, 0.78 at 0.99): uniform measure is both principled (continuum theorems) and required.

Leg-shedding kilonovae. $M_{\text{ej}} = \Delta k m_{\text{leg}} \times 0.1$, blue ($v \sim 0.3c$, $t_{\text{peak}} \sim 0.6\text{--}1$ d) + red ($v \sim 0.1c$); $L \propto t^{-1.3}$ tail. Band mags assume BC = 0 and per-component peaks; POSSIS refinement queued. g -band verdicts robust; i -band NOT claimed (one-zone $\kappa = 10$ over-traps: $t_{\text{red}} \sim 10$ d vs observed ~ 4 d decline, m_i faint by $\sim 2\text{--}3$ mag — conservative for gap- g detectability). Standard 2–28% band is a literature EOS input, not derived.

O5 protocol. Trigger: $m_1 \in [2.5, 5.0] M_{\odot}$, < 200 Mpc, 90% area $< 100 \text{ deg}^2$ (multi-detector); Rubin ToO epochs 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Kill rule: 10 qualifying non-detections kill BU. Archival ledger: GW230529 uninformative (single-detector, 93% unsearched); S250206dm unverified (check GCNs before citing). Full plan: docs/observation-protocol.md. Figs. 66–69.

Mass-gap audit (module massgaps). Lower gap: $k(M)$ log-log curvature $< 10^{-9}$ over $0.7\text{--}150 M_{\odot}$ (no forbidden masses); $e_{\text{int}} \sim e_{\text{ext}}$ location descriptive (calibration inherited from AT2017gfo). Upper gap: spin-leg ratio $1 \rightarrow 0.857 \rightarrow 0.5$ at $a = 0 \rightarrow 0.7 \rightarrow 1$ with no jump on $[0, 0.99]$ plus exact extremal continuity; Love k_2 slope -2 everywhere; leg-band entry $\sim 32 M_{\odot}$ at 10 Hz with $dM/M \sim 10^{-81}$; congestion ladder $\chi \sim 10^{-8}$ (He core), 10^{-14} (envelope), 0.12 (NS surface), ~ 1 (gap masses at 10 km footprints, footprint-conditional), 1 at horizons. Shedding ex-

Fig 74 — D2: unitary evaporation ($V^d V = I$, S from ρ_{rad})

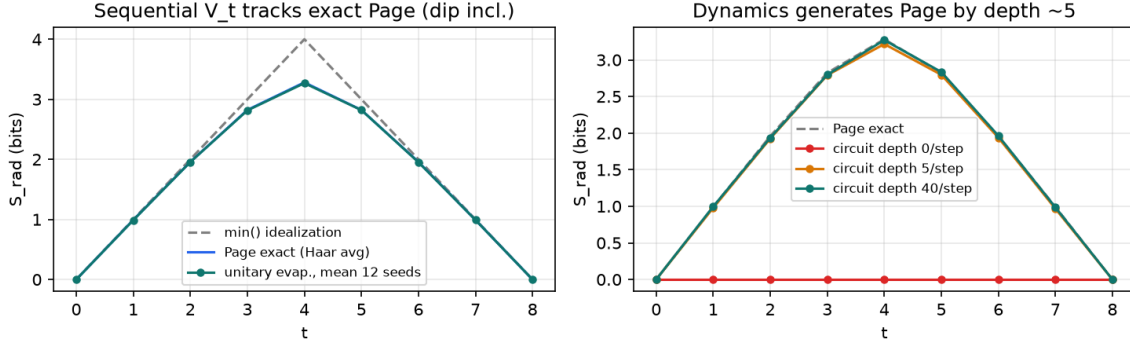


Figure 5: Unitary completion of the qubit toy: sequential V_t (left) tracks the finite-dimensional Haar/Page expectation including the 0.72-bit dip, computed from ρ_{rad} rather than $\min()$; finite-depth all:all circuits (right) converge to Page by depth $\sim 5/\text{step}$ without assuming Haar. Module `evaporation_unitary` (D2).

Fig 66 — BU: 2PN cancellation at $p=0.92$ (J0737 0.00 sigma)

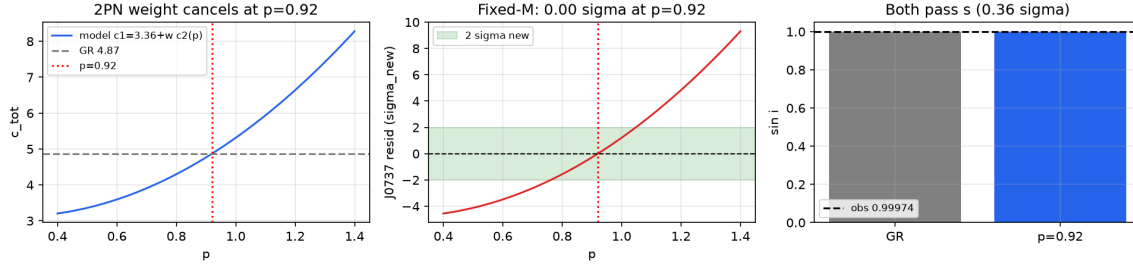


Figure 6: 2PN cancellation: model $c_{\text{tot}}(p)$ crosses GR at $p = 0.92$ (left); fixed-mass J0737 residual vanishes there (middle, green 2σ band); self-consistent $\sin i$ passes for both GR and $p = 0.92$ (right). pulsar.

actly flat in q (machine precision) — but q -independence is prescription (shed fraction $\times$ efficiency fixed once on AT2017gfo), not derived from the graph-dynamics evaporation isometry V_k (open, S1/S3/D1): $q \sim 0.11$ (GW190814) flashing at the same fraction as $q = 1$ (GW170817) is the model's biggest theory bet. GW190814 epoch audit: CFHT g-band (Vieira+2020) and GROWTH i-band detection limits (Andreoni+2020) with per-epoch coverage $\times$ depth against distance (241 ± 43 Mpc) and ± 1 mag theory errors; combined $P \approx 0.68$ (g-only) / 0.88 (g+i at $g-i = 0.7$). Erratum on record: the first audit used a nominal ZTF depth, but ZTF ran no targeted GW190814 follow-up; superseded by the CFHT/GROWTH epoch tables above. Fig. 75 (main text). Analytic systematics (round 4): viewing $A_g \sin^2 \theta$ ($A_g = 1.25$) + opacity-rescaled blue mags give the P grid in Fig. 75b; the hiding window is bounded, not open-ended.

S5 UV nulls and phenomenology (T, U, V, BD–BF)

TeV: $M_D = 1$ TeV gives $k \approx 11$ (3 TeV) – 17 (13 TeV) vs $k_{\text{crit}} \approx 50$ — thermal nulls expected, onset ~ 550 TeV. Echoes: $\Delta t = 4M \log(M/l_p)$, 0.03–0.3 s (GW150914 0.11 s), in searched windows but $R \sim 10^{-160}$ unobservable. EHT: $l_p/2M \approx 10^{-48}$, 47 orders below sensitivity. Dispersion:

Fig 67 — BU: gradient OR gives $p=0.92$ (exact EMD)

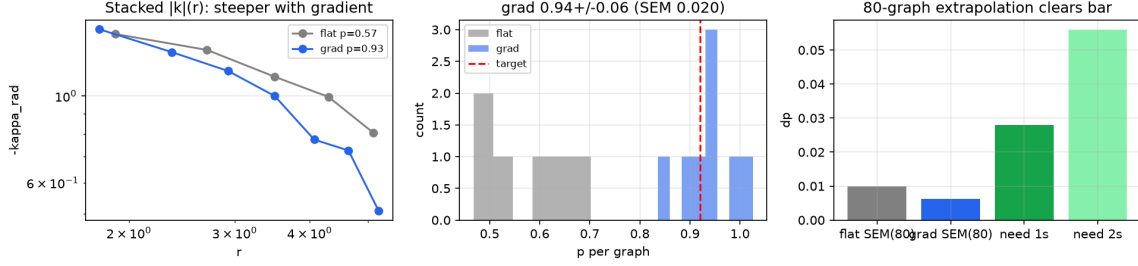


Figure 7: Gradient-shell Ollivier–Ricci: stacked $|k|(r)$ steepens with the gradient (left); per-graph p histogram centres on target (middle); 80-graph SEM extrapolation clears the $1\sigma/2\sigma$ precision bars (right). Production $N=1020$: $p = 0.913 \pm 0.049$, SEM 0.0055. **orici**.

$E_{QG,2} = \sqrt{8} E_P$, Fermi-safe by 10^8 ; universal GW extrapolation shifts 100 Hz by $\sim 10^{-82}$ (LVK $\alpha = 4$, safe by $\sim 10^{60}$ — a recorded null, not a falsifier; no GW sector claimed). Chromatic lensing 10^{-56} – 10^{-32} . Figs. 21–22, 39, 51–56.

Fig 51 — BD: quadratic lattice dispersion, Fermi-safe

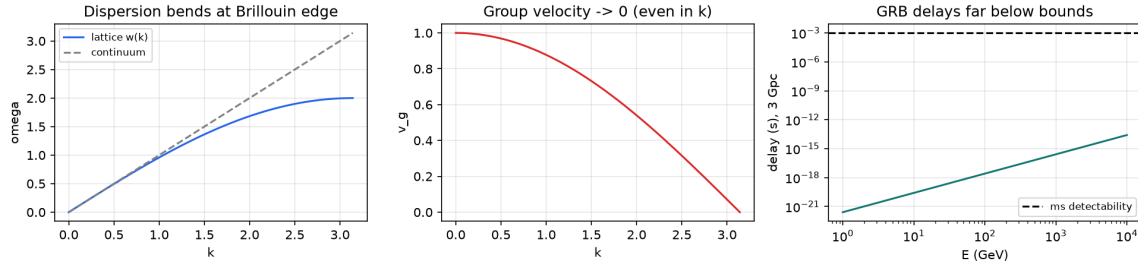


Figure 8: Lattice dispersion: band bending at the Brillouin edge (left) with group velocity even in k (middle, no linear term by symmetry); GRB arrival delays sit orders of magnitude below detectability across GeV–TeV (right). **dispersion**.

S6 Kill list (AN + BU)

AF quench ratio < 1.3 ; α outside $[9.0, 12.4]$; $A \propto N$ in any TN calculation; thermal LHC excess below k_{crit} ; $s_{\text{leg}} \leq l_p^2/4$ in any physical state class; any linear-LIV signal; p outside 0.92 ± 0.056 at $N = 1024$; graph-predicted $|\delta_Q| \gtrsim 0.17$; simulator d_{eff} shape-mismatch after the cost mapping is fixed (D10/P0' wire); 10 gap non-detections; O5 nearby-BBH ToO sample clean (universal shedding, Eq. 2 main). Field-program wires (S12): a second admissible M_O with robustly non-3D IR refutes reconstruction universality; $I > 0$ in any collision cell beats the linearity null; a firing trigger derived from the instantaneous ontology overturns BR27-NO-MODE (strong stop: no rate shopping; TRIGGER-0 census: 19 conditions, 0 implications); an earned local covariant rewire selector overturns REWIRE0-DEGENERATE; a sub-minimal store (cover-only or scalar- R) recovering exact predecessors overturns STORE0 minimality; any inter-event update of the stored record Q under fixed-graph waiting overturns QDYN0B-EVENT-LOCAL; a MAX area law without the complete core, or $h^* < 1$ on headline geometry, overturns BHQAREA0 core-selection; a genuine jet surface under unchanged definitions overturns JET1-NULL (stop rule binds until then); a ten-

sion geometry with no U-shape under the fixed cost rule, or amplitude anti-scaling in χ , kills the tension-imprint conjecture. One sub-claim (broad remnant DM) already ruled out on the record. Figs. 37–39.

S7 N-scaling and calibration record

Drift check: $\beta = 1.5$ calibrated at $N = 300$ drifts high at $N = 600$ (1.16 ± 0.06); recalibration to 1.28/1.24 recovers $p \approx 0.92$. Pre-run $1/N$ extrapolation predicted $\beta(16000) \approx 1.18$; measured 0.74 — drift faster than $1/N$, now fit log-linear. $N=1020$ cross-check: CSR-direct 0.9107 vs NetworkX 0.9134 (agree to 3×10^{-3}). Local- p turnover (inner windows flatten, 0.63–0.69 $\rightarrow$ 1.26–1.46) is measured at 4k/8k/16k, not extrapolated. Fig. 73.

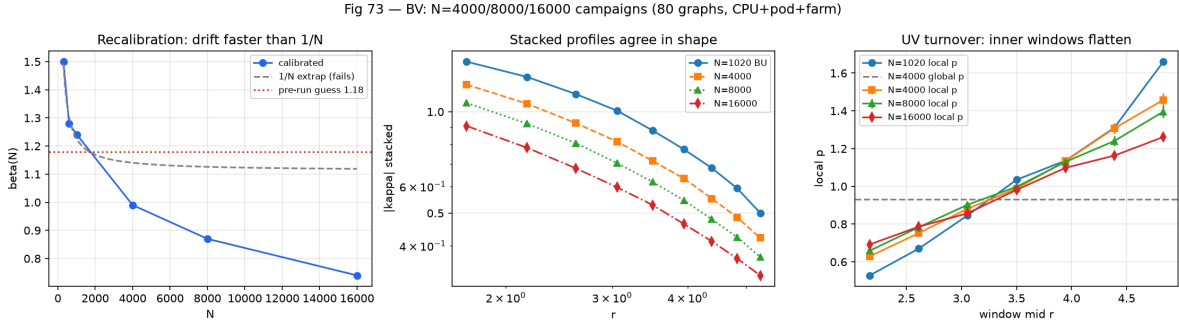


Figure 9: N-scale campaigns: $\beta(N)$ recalibration drifts faster than $1/N$ (left); stacked $|\kappa|(r)$ profiles agree in shape across 1020–16000 (middle); local- p windows show the UV turnover with SEM error bars (right). 80 graphs per production row. **shellscale**, **sinkor**.

Reproduce ($N=1020$, ~ 9 min):

```
measure_p(per_shell=102,n_shells=10,n_graphs=80,
gradient=True,beta=1.24,seed0=0,max_per_shell=8)
```

S8 PPN ledger and walk/foam/healing notes

Walk no-go (BP/BQ). All smooth weight rules give drift slope -2.99 ; only the absurd $e^{2\sqrt{d-d_0}}$ escapes. Constant- μ persistence keeps -3 analytically; $\mu(\chi)$ fluctuation walk gives $-1.95 \approx -2$ modulo the labelled $\sqrt{\chi}$ assumption. Harmonic/hitting-probability walks fail by leg shorting. Conclusion: temperature (AS), not bare graph diffusion, carries Newton. Figs. 63–64.

Foamgrid FDTD (BF). Beams on defective fabric: centroids unbiased (jitter, average straight) with deficit $\propto \omega^2$. Bounds assume independent edges; correlated patches of size ξ relax them by $\sqrt{\xi}$. Fig. 56.

Healing ladder (AG/AH). $\tau_{\text{heal}} = 11.24M$ ($1/\text{Im } \omega_{220}$); GW150914-mass ladder: healing 3.5 ms, scrambling 0.23 s, Page $\sim 10^{79}$ s, evaporation $\sim 10^{80}$ s. MSS $\lambda/2\pi T = 0.50$ – 0.68 ; LVK $\alpha \in [9.0, 12.4]$ contains 11.24. Figs. 30–31.

S9 Archival ledger and O5 trigger (BU detail)

Trigger (full plan docs/observation-protocol.md): $m_1 \in [2.5, 5.0] M_\odot$, < 200 Mpc, 90% $< 100 \text{ deg}^2$; Rubin ToO 0.5/2/5 d in $g+r+i+z$ (DECam/Gemini backup). Sought: fast blue ($v \sim 0.3c$,

Fig 30 — AG: healing lag + timescale ladder

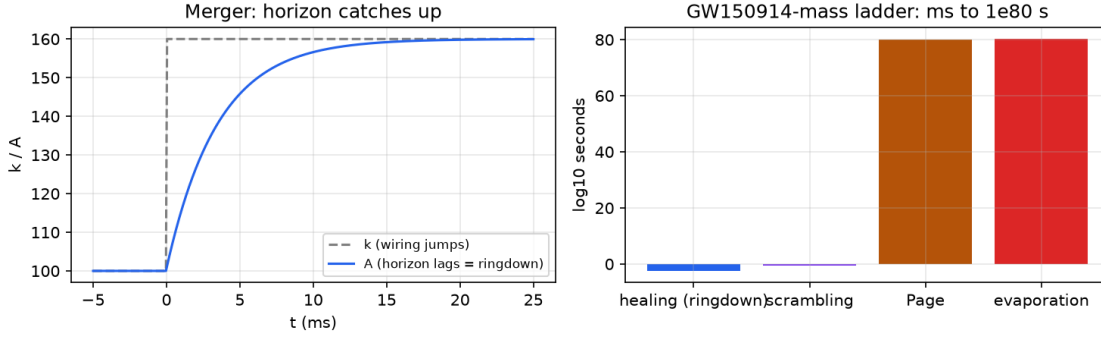


Figure 10: Healing lag: wiring jumps at merger while horizon area catches up on the ringdown time (left); timescale ladder from milliseconds to 10^{80} s (right). The final sigh carries $\sim 10^{-78}$ of the mass. healing.

$t_{\text{peak}} \sim 0.6\text{--}1$ d) $\rightarrow$ slow red with $L \propto t^{-1.3}$ tail. Yield: $1.5 \text{ gap/yr} \times 70\% \approx 1.05/\text{yr}$ ours vs $\leq 0.3/\text{yr}$ standard. Yield arithmetic in `collapse.gap_o5_yield`; brightness in `collapse.peak_apparent_mags`; GW190814 epoch audit + systematics grid in `massgaps.gw190814_combined_pdetect / massgaps.gw190814_systematics_table`. Figs. 68–69, 75 (main text), 75b.

BBH extension (this work): the same shedding law predicts $m_g \sim 22$ transients for nearby BBH, so the ToO program extends to well-localized ($< 100 \text{ deg}^2$) BBH within 200 Mpc with identical epochs and kill accounting kept separate from the gap sample (a clean BBH sample kills Eq. 2 main even if gap events flash). Full thresholds in `docs/observation-protocol.md`.

Fig 69 — BU: O5 detectability + GR-battery 2PN margins

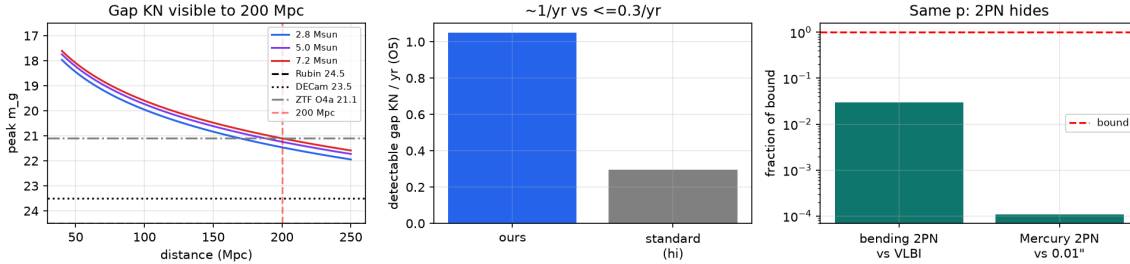


Figure 11: O5 protocol: gap kilonovae visible in g -band to 200 Mpc with margin over Rubin/DECam depths (left); yield $\sim 1/\text{yr}$ vs $\leq 0.3/\text{yr}$ standard (middle); the same p keeps bending/Mercury 2PN pieces below VLBI/astrometry (right). collapse, pulsar.

S10 Module map and reproduction

Modules follow the appendix letters of the retired v4.1 record: `graphs`, `scrambling`, `emergent_dim` (L0 dimensions), `horizon`, `micro`, `circuits` (A), `maxent` (B), `qes` (C), `evaporation` (D), `evaporation_unitary` (D2), `qec` (F), `robustness` (G), `kerr` (H), `thermo` (H2), `haar` (I), `monogamy` (J), `otoc+pheno` (L), `tn` (M), `kerrpage` (N), `syk` (O), `data` (Q), `litcompare` (R), `tev` (T), `echoes` (U), `posteriors` (W), `ds` (X), `krylov` (Y), `collapse` (AA), `cosmic` (AB), `lunch+remnant` (AC), `bounds` (AE), `healing` (AG), `mss` (AH), `bigsyk` (AI), `mp+greybody` (AJ), `congestion` (AK), `charge` (AL), `bandwidth` (AM), `gridcirc+monitor+selfattack+lhc` (AN), `concentration` (AO), `ps` (AP), `scatter` (AQ), `emd+viability` (AR), `entropic` (AS), `redshift` (AT), `heatker+orici+jacobson`

Fig 75b — GW190814 systematics: equatorial + kap=2 hides (P~0.18)

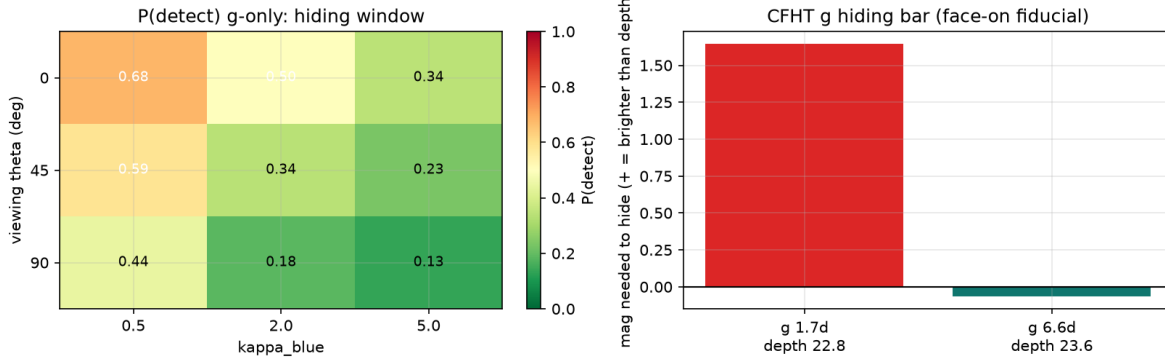


Figure 12: GW190814 analytic systematics (round 4; full POSSIS queued, D7). Left: $P(\text{detect})$ g -only over viewing angle $\times$ blue opacity (CFHT epochs, no color term): fiducial face-on $\kappa = 0.5$ gives 0.68; equatorial + lanthanide-mixed ($\kappa_{\text{blue}} = 2$) drops to ~ 0.18 . Right: face-on hiding bar per CFHT g epoch (g 1.7 d needs ~ 1.6 mag). Module `massgaps`.

(AU), `fission` (AV), `klanguage` (AW), `tension` (AX), `gw250114` (AY), `overtones+tensionvol` (AZ), `sparse24` (BA), `lensing+chroma+shapiro` (BB), `bcrit` (BC), `dispersion` (BD), `qnmfoot+qnmlegs` (BE), `foamgrid` (BF), `perwalk` (BG), `strain` (BH,BJ), `weakfield` (BI,BQ), `leggham` (BL), `pulsar+orici` gradient shells+collapse leg-shedding (BU), `uvscatter+sinkor+shellscale` (BV), `massgaps` (mass-gap null + GW190814 epoch audit). Update-rule and dimension programs: `update_rule` (D1 tournament), `blind_u` (D1 blind- U), `weighted` (D10/D14 pricing), `spectral` (D10 second leg), `formation` (D14 C2/SSB-1/Stage-0). Field program (S12): `ballistic` (P1 waves), `coherence` + `slit` + `tunnel` (interference), `potential` (POT-0) + `driven` (POT-1), `continuum` (EM-0) + `falsification` (EM-1), `malus` (sheet sectors), `obs0` + `obs0r` + `obs1` + `obs1_reveal` + `quot` (rulers, blind observer, quotient), `backreaction` (BR-0) + `phase` (BR-2) + `rigidity` (BR-1), `contraction` (BR-2.5) + `conservation` + `accounting` (CONS-0/BR-2.6) + `stability` (BR-2.7), `u0` + `time0` + `rand0` + `measure0` + `sym0` (dynamics completion + state census), `field0` (linearity null), `response` + `bgresp` (kernel + vacuum susceptibility), `vac0` + `vacfield` + `vacexc` + `vaccomp` + `vacselect` (Final MIXED + vacuum family), `hidden` + `hiddenbr` (dead-sector info + response), `zero` (nodal census), `grav0` (graph-only null), `split0` + `rewire0` (inverse fiber + rewire null), `merge0` + `reservoir0` + `trigger0` (update + account + conditions), `fiber0` + `info0` (fiber debt + exact books), `vacdomain` + `vactexture` + `vacstab` (vacuum trio), `source0` (boundary data), `store0` (reversible store), `dim3` + `dim3_reveal` (3D lift + blind reveal), `subclass0` + `scale0` (substrate class + large- L bank), `qdyn0` + `qdyn0b` (store-dynamics null + event-local readout), `bhent` (collapsed census), `dim31` (repaired 3D rulers), `weave0` (random-weave core search), `qinfo0` (store/qubit identity), `bhqent0` (blind-store dimension), `bhqarea0` (boundary Q area law), `jet0` + `jet1` (jet search + NULL repair), `event0` (event necessity). Side apparatus: `spectroscopy` (SPEC0 null), `stern_gerlach` (SG gate fails), `fep` (FEP0 null), `mergershed` (BU2 q -shape derived, flat law kept as O5 falsifier), `graphvk` (D1 graph instance closed at small- N ED).

Reproduce:

```
pipinstall-e".[dev]"
pytesttests/-q
pythonscripts/generate_figures.py
streamlitrunapp.py
```

S11 Emergent-dimension methods (L0)

Protocol. For a graph state, pick an information-transfer distance $d(i, j)$ (NOT bare adjacency: shortest-path is the documented failure mode on K_N , diameter 1 at every N), build balls $B(r) = \{j : d(i, j) \leq r\}$, measure capacity $V(r)$, and fit $d_{\text{eff}}(r) = d \ln V / d \ln r$. Observed dimension is the infrared conjecture $d_{\text{obs}} = \lim_{r \rightarrow \text{IR}} d_{\text{eff}}(r) \simeq 3$ (module `emergent.dim`; 12 tests). Total

emergence requires the two-distance bracket: shortest-path balls (adjacency scaling) and diffusion balls (information-spreading scaling) must agree at 3 on the same graph with nothing dimensional imposed.

Ruler validation (imposed lattices, circular for emergence). Shortest-path converges from below (exact BFS shells): chain $0.92 \rightarrow 0.96 \rightarrow 0.98$ ($N = 60/200/600$), 2D $1.67 \rightarrow 1.77 \rightarrow 1.82 \rightarrow 1.87$ ($L = 20/40/60/100$), 3D $2.25 \rightarrow 2.46 \rightarrow 2.56 \rightarrow 2.63$ ($L = 7/11/15/21$), gaps $\sim 1/\sqrt{L}$ (Manhattan-diamond shape + open box). Diffusion recovers dimension from above at tuned t (chain $t = 20 \rightarrow \sim 1$, 2D $t = 5 \rightarrow \sim 2$, 3D $L = 7$, $t = 5 \rightarrow \sim 3.3$); the 3D overshoot shrinks with size ($3.56 \rightarrow 3.34 \rightarrow 3.14$ at $L = 5/7/9$, $t = 5$). Degree-3/4/6 lattices all scale 2D: dimension comes from 2D-ness, not coordination.

Rejections (negative pins). Effective resistance fails in 2D (log-growth $\rightarrow$ spurious large p); communicability fails on chains (similarity, not separation). Radial shells fail both sides (shortest ~ 1.4 , diffusion ~ 0.6): radial-only wiring gives ~ 1 D with no 2-sphere factor — the missing-area diagnosis that motivates P0'. On K_N , shortest is trivial and diffusion uniform, by design (T1: no interior geometry).

Tense plug (D10a). A 5×5 clique plug at the centre of a 40×40 grid: tense-centre mid-window ($r = 8\text{--}20$) scaling dips below fabric (~ 1.52 vs ~ 1.89 control) and recovers from below — *inverted* vs the GR far side (above 3). Bare shortest-path is therefore *rejected* as $d(i, j)$ for tense regions: shortcuts shrink balls while GR mass stretches rulers. Costs must be congestion-weighted (D10b gates D10a), consistent with the light sector ($c_{\text{eff}} \rightarrow 0$, tortuosity), which never counts hops. Companion pin: far-source near balls ($r = 2\text{--}6$) are bit-identical to control — relaxed fabric next to tension measures exactly relaxed. Far-field tail (D11, closed): $\chi \sim 1$ decays $1/r^2$ ($E \cdot r^2$ flat to $10R_c$), $\chi \sim 2$ decays $1/r$ ($E \cdot r$ flat to $10R_c$), $\chi \sim 5$ slow crossover (q 1.19 $\rightarrow$ 1.10 to $20R_c$; $1/r$ supported, not firm) — the exponent is tension-dependent. κ -profile (D10b, closed): corner-free disk plug reproduces the boundary-negative/core-positive pattern quantitatively ($+0.90/-0.86/0.00$ vs clique $+0.89/-0.93/\sim 0$); mild-disk matches mild-square ($-0.05/-0.23/0.00$).

GR fingerprint (D10 target, computed). Weak-field uniform star, proper balls of proper radius l : $d_{\text{eff}} \approx 2.990$ at $l = 30M$ (near dip), 3.030 at $100M$ (overshoot), 3.010 at $1000M$, 3.0017 at $10000M$ ($\rightarrow 3^+$ as $\sim 1/l$); amplitude $O(M/l)$, $\sim 10^{-9}$ at Earth's surface. The far-field overshoot is structural: a localised mass leaves $V(l) = \text{flat} - \text{deficit}$, so V/V_{flat} rises toward 1, which *is* $d_{\text{eff}} > 3$. Close criterion (D10a): a tensed simulator region reproduces this *shape* under info-side d_{eff} once the cost mapping is fixed; shape failure refutes P0'. Reproduce: `pytesttests/test_emergent_dim.py-q`.

Reconstruction universality (D12, open). The observer map M_O is not unique on its face: closeness, resistance, diffusion, and communicability distances are all candidate reconstructions. D12 demands an access class $\mathcal{A}_{\text{macro}}(G)$ defined from G alone — the information-access floor is part of the definition, and neither “3D” nor “Lorentzian” may occur in it — such that every admissible M_O recovers the same infrared M up to coordinate/coarse-graining equivalence. Falsifier: a second admissible M_O giving robustly non-3D IR on relaxed fabric refutes the P0' reconstruction program. Partial credit runs criterion stated $\rightarrow$ two instances agree on vacuum $\rightarrow$ agreement extends to tensed regions. P4 (uniform Ollivier) audited against the five: criteria 1, 2, 3, 5 pass (permutation-covariance and drift-localization tested, the latter on flat and plug backgrounds), 4 partial. Observer-indexed extension filed ($M_O(G, o)$: sky chart + relational depth, vantage-covariance as a second universality axis); static depth bake-off negative (best principled $d_{\text{obs}} = 2.28$ vs tuned-null 2.73; MDS readout likewise negative, logC gap shrinking with N). Adopted reframing: holographic DOFs + dynamical relational distance $\rightarrow$ emergent localization, M_O demoted to readout; rank-3 claim needs an N -stable λ_3/λ_4 gap with non-radial v_3 . Tier-1 MDS calibration pinned: lattice/gabriel/kNN hop show λ_1, λ_2 dominance, while Delaunay hop carries

a radial tortuosity bowl ($v_3 \leftrightarrow r^2$ at 0.93) – the MDS bar is per-substrate, and eigenvector geography is part of the test. Shell-counting dilemma sketched: honest per-shell counts on 2D fabric (nodes $\sim r$, cut $8r+4$, MDS rank $O(1)$ with $\text{GoF}_2 = 0.772$) never reach r^2 , while cumulative-ball r^2 violates shell-independence by construction – $d_H=2$ XOR independence, with non-geometric C_O or non-shell composition left as escapes. The dilemma binds hop shells only: the MDS-rank route is independent, and dynamical metrics redefine shells (first shipped dynamical distance: SI first-passage, MDS null cleanly 2-dominant with $\lambda_2/\lambda_3 \approx 17.5$).

Staged causal order (D13). Time undergoes reconstruction exactly as space does: the microscopic update index n , the causal order $\prec_U$, and clock time τ are three distinct orderings. The static control (D13.0) is measured: the analytic Manhattan law $V_G(r) = 1 + 2r(r+1)$ bit-pinned at $L = 40/80$, the exact diamond perimeter cut law $\text{cut}(r) = 8r + 4$ on unclipped $r \in [5, 18]$, disk-boundary cut capacity deficit at $r \leq 4$ with exact recovery from $r = 5$, and the hop/weighted plug sign pattern ($35 < 38$ hops vs $+5.0$ weighted delay) pre-registered for dynamical confirmation. Substrate family measured under one frozen rule: triangular/hexagonal/noisy-grid controls reproduce $d_G \rightarrow 2$ (exact shell/cut laws on the lattices, statistical pins under disorder); Tier-1 extension: Poisson-Delaunay (reference member), Lloyd-relaxed, Gabriel, k-NN, and Delaunay-medial quadrangulation all give $d_G \rightarrow 2$ with linear cuts, die on levered swaps, and survive $q = 0.05$ deletion, while span-limited swaps preserve 2D-ness — degeneracy across topology/degree/order is the expected universality signature, and the physical object is the class $[G]_{\sim_O}$, not a winning tessellation. Spectral dimension is the second leg: torus heat-trace $d_s(t)$ holds $(1.95, 2.05)$ over $t \in [10, 100]$, and Weyl counting fits land every member in $(1.90, 2.15)$. Exact-UIPQ sampling stays queued (free-walk labels are not conditioned well-labeled trees); Lloyd is excluded from spectral comparison (unclipped relaxation coalesces points above $N \approx 1600$). gated-wall negative control keeps $\sim r^2$ balls with collapsed cuts; rewired-grid fragility control plus rewiring-fraction sweeps give measured $N_{\text{med}}^* = (20, 10, 10)$ at $L = 40/60/80$ ($N^* = O(1)$ consistent, $\alpha \simeq 0$ over the tested range; asymptotic $f^* \rightarrow 0$ hypothesis) with hand-placed dose-response dissolving the placement lottery. RG blocking (2×2 , $L40 \rightarrow 5$) flows rewired long-edge density $0.013 \rightarrow 0.31$ against plain 0.0 (rough $y_\lambda \approx 1.5$). Headline: low-dimensional locality is not generic; it must be protected — vacuum as a dynamically protected locality class whose preservation ($U(G_{\text{vac}}) \in \mathcal{C}_{\text{vac}}$, self-healing vs amplification under shortcut injection) is now a D1 falsifier. BFS first-passage is hop distance, so this control is static geometry, not propagation. Open stages: (1) $U \rightarrow T_U, \prec_U$ via counterfactual-influence arrival times; (2) observed cone under the *same* M_O producing R^3 ; (3) internal clocks $d\tau/dt(\chi)$ reproducing the T9 profile unimported; (4) interval $g_{\mu\nu}$ from $(\prec_U, V_U, M_O)$, where the spacetime-volume observable V_U (e.g. calibrated event counts in causal diamonds) is independently defined — spatial $V(r)$ is reachability, not event volume. Malament’s theorem is motivation conditional on the continuum correspondence (order fixes the conformal class; V_U the factor), not an earned mapping. Companion constraint on U (D1, first pass): six update rules tournamented against shortcut damage all stall with residual longs (best single-move: slide rule, $30 \rightarrow 5$) while p -healing proves gameable (greedy perfects p to 10^{-6} while tripling longs) — judged on (p, longs) jointly. Second pass: coordinated double-swap breaks the floor ($30 \rightarrow 1$, pair-locked residual) and the visibility-chained triple endgame fully heals (longs $\rightarrow 0$, two seeds plus torus $2/2$; triple-lock localizes to boundaries) — locality protection demands coordination order 3 with detour-aware proposals, order 4 with a strict census gate for corners (ungated order-4 scrambles: $10/10$ accepts while longs $1 \rightarrow 5$). Gate duality filed and split by a temperature knob: strict order-3 clears dm4 in 0.5 s but stalls seed7, $T_0=2$ annealing clears seed7 — neither pole dominates (stall topology differs). The knob replicates at order 2 (pair_anneal matches-or-beats pair on every stream); gate-default adoption rejected (worse-on-s5 default is bad science). Reproduce: `pytesttests/test_scrambling.pytests/test_substrate_family.pytests/test_spectral.py-q` (D13); `pytesttests/test_update_rule.pytests/test_`

blind_u.pytests/test_weighted.pytests/test_j2.py-q (D1/D14/J2).

Dynamics coherence (C1–C5, over-arching). $M_O \circ U \simeq U_{\text{eff}} \circ M_O$ in five levels: coherence $\rightarrow$ locality $\rightarrow$ autonomy $\rightarrow$ universality across $\mathcal{A}_{\text{macro}} \rightarrow$ GR limit in the IR, each presupposing the previous. Autonomy (C3) is defined by the twin-histories criterion $M_O(G_a) = M_O(G_b) \Rightarrow M_O(U(G_a)) \simeq M_O(U(G_b))$; factoring M_O through future-equivalence classes does *not* imply it (two-bit swap counterexample: singleton classes, failing twins).

Update-rule tournament, third pass (D1). Blind- U reframing adopted: stationary-ensemble vacuum with U -admissibility (drift / square / triangle first entrants; Metropolis no-window, grid-not-optimum, and kappa frozen-descent negatives; triangle-landing misdirection plus degree-fiber reachability). Cross-candidate battery opened (triangular healing generalizes $24 \rightarrow 13$; hex needs fabric-relative radius). Coordination need is substrate-dependent: triangular pair alone fully heals ($24 \rightarrow 0$, p bit-plain); hex floor resolved (leaf edges, interior census 0, guillotine $54 \rightarrow 17$); Delaunay reference heals at order 2 (pair600 $21 \rightarrow 0$); Gabriel hits a broad-spectrum census limit (descent overshoots below plain); medial mixed (strict gate $6 \rightarrow 0$ in one accept); k -NN mostly-heals (floor 16); Lloyd heals via chain (pair grinds to 1, strict clears). Self-calibrating census (median-relative; exact on gapped, Gabriel unseparated). Reproduce: pytesttests/test_update_rule.pytests/test_blind_u.py-q.

Pricing-hierarchy origin (D14). Weighted-graph audit adopted (event sweep + edge census; 3-vertex cross-checks; weight-tolerance recovery at $L_w=20$; T15 cites Prop. 1). Weighted RG: $L_w=10$ washes out by level 3 (pricing irrelevant above span scale; vacuum RG-protected, no pumping needed); Φ -apparatus with tolerance contour; knot pilot (dual ruler, no bundle mixing, ruler-trap law). Experiment A (Form-1 relaxation): control bit-exact, $\beta=350$ recovers the banked slice ((36, 23, 7, 0) vs (36, 23, 5, 0)), Δy_w split (substitution at price level, span-selection at Φ level). Walk-atrophy killed (0/8 horizons plus inverted signal; avoidance is conductance math). Betw-congestion closure: feedback attenuates ($m=1$: factor 0.684, in-basin cross at $k=3$; $m=4$ exit is a cadence artifact), concave- D holds — static χ closed. Stateful-scale derivation: linear memory equals static (LTI steady map), nonlinearity buys shape not magnitude (characteristic-scale lemma), YES-prong criterion (gain must be a structural number; unoccupied). Gain-free discriminator fails (weak median 6.2, rows (39, 30, 20, 4), no crossing; threshold narrowed to (6.2, 11.0)): the gain is debt by elimination, and magnitude origin passes to formation/dynamic topology. Reproduce: pytesttests/test_weighted.py-q.

D14 formation design (C2 machine). E-conserving relocation on ER/RR soup: C2-PILOT-1 topological verdict DARK (54/54, 0/54 WEAK over a $20 \times$ floor sweep; D1 null holds as Poisson equilibrium; D3 hope dead — 13 dust-shedding arrests + 23 stillborns, zero active; no concentration pathway, so the D5 triangle-closure lead). C2-PILOT-2 geometric verdict NO-coexistence (66 + 6-repair runs, bit-identical cross-machine): CONDENSATION 9 (D5 κ_2 @1600 single-clique) vs FRUSTRATED 15 (D5 ∞ all- N , kmax-4 churn limit, N -independent, active-7% EXCHANGE) vs POISSON+ 39; $\kappa_c \in (1, 2)$, nucleation window $N^* \in (1600, 3600]$, kinetics mirror-broken, refined D3 ... D5 ∞ ... D5 κ_2 bracket (frustration-as-selector lead). SSB-1: core selection SPONTANEOUS (same-soup Jaccard $\approx$ random), so D15 reopens with symmetry restated as statistical- S_N -of-law. Stage-0: blob kinematics NULL (0/6 directed, 0/6 persistent-handed, achiral plateau) — the debt-free pure-D5 ∞ route stops and polarity pauses (individuals yes, circulation-sign no). Reproduce: pytesttests/test_formation.pytests/test_closure.py-q.

J2 micro/macro probe (D10). $J_2 = \mathbb{Z}^2 \rtimes \mathbb{Z}_2$ walk graph (degree 8; substrate of D’Ariano–Erba–Perinotti, Phys. Rev. A 100, 012106 (2019)), pre-registered as a probe, not Tier-1-track (index 2 over $\mathbb{Z}^2$, hence quasi-isometric to the square lattice: long-scale $d=2$ is a control). Quotient verified computationally (1861 cells, every edge square-adjacent, 1741 interior cells square-4 with micro-multiplicity 4, quotient shells exactly $4r$); exact micro laws (shells $8r$ for $r \geq 3$, cuts

$32r+16$ for $r \geq 2$); bipartite correction filed ($q=x+y$ bipartition, odd returns vanish exactly — a pre-registered aperiodic prediction falsified with mechanism); 4-cycle census 26072 at $R=18$ ($\sim 20 \times$ square-equivalent); perturbation family-typical (swaps $2/3$ kill, q -deletion robust) — no tier claim. Orientation program (D14/D15, 32 runs on the J2 torus): cores form despite the pristine triangle-free substrate (kmax-4, triangle-free bootstrap), radial SSB preserved, but orientation absent (random-axis elongation, edge $x/y \leq 8\%$, quadrant signs null-consistent, sheet 0.49–0.50) — verdict (c)-ISOTROPIC, so D5 ∞ -radial is the wrong kind for compass consumption (D15 stays closed on this candidate). J2 is adopted as the canonical working vacuum substrate (docs/j2-status.md; evidence-based, non-uniqueness explicit). Reproduce: `pytesttests/test_j2.py-q`.

S12 Relational field program (F-layer)

Postulates and anatomy. FP1: two real scalars per node, $\psi_u = r_u + is_u$ (no vector, momentum, or coordinate register). FP2: $i\partial_t\psi = H\psi$ with $H = -A$ (frozen $J = 1$), the one-way $G \rightarrow \psi$ map, postulated as the working law. Granted only this, VAC-0A derives the anatomy for every simple undirected graph (A1–A6 identities on hostile graphs $\rightarrow$ LAW): conserved $\rho_u = |\psi_u|^2$, $Q_\psi = \sum_u |\psi_u|^2$; bond energy $B_{uv} = \text{Re}(\psi_u^* \psi_v)$; continuity current $J_{u \rightarrow v} = 2 \text{Im}(\psi_u^* \psi_v)$ with $d\rho_u/dt + \sum_v A_{uv} J_{u \rightarrow v} = 0$; energy $E_\psi = \langle \psi | H | \psi \rangle = -2 \sum_{(uv) \in E} B_{uv}$ and the conjugacy $\partial E_\psi / \partial A_{uv} = -2B_{uv}$ (first pinned on the J2 stack inside EM-0). Quadrature $B \sim \cos \Delta\theta$, $J/2 \sim \sin \Delta\theta$; some readouts store the bare imaginary part (exactly half the current — convention, not a second current). Reproduce: `pytesttests/test_vac0a_identities.pytests/test_vac0_battery.py-q`.

Propagation, response, rejected EM reading. POT-0 separates direction from any node arrow (POT0-COLLECTIVE: symmetric source spreads, coherent packet goes ballistic, scrambling kills direction, coherence restores it). POT-1 + EM-0 give one field with a static regime and a propagating front on source change (EM-0 continuum reading: J2 Bloch bands, long-wave Schrödinger sector, continuity to numerical zero; no Maxwell/charge/photon claim). EM-1 rejects the electromagnetic identification (EM1-FALSIFIED): static range saturates ($\xi \approx 0.53$, range 3), one propagating scalar mode, local phase moves $B/J/E$ by order one, nodal drift tracks $v \cdot q$ not $v|q|$; the conserved signed charge is sheet-tied with frozen conjugate. Reproduce:

```
pytesttests/test_potential.pytests/test_driven.py-q
pytesttests/test_continuum.pytests/test_falsification.py-q
```

Interference apparatus: two-path PASS on open square, square torus, J2, and the J2 quotient; FAIL on tri/hex; rewires fail the phase rung; MZ corridor LAW-level after a window erratum (VAC-0F completion: square class, not J2-only; H/I closed below, VAC-0 FINAL MIXED). Reproduce:

```
pytesttests/test_slit.pytests/test_coherence.py-q
pytesttests/test_tunnel.py-q
```

Operational geometry and quotient. OBS-0R meets six rulers on J2 (balls/Hausdorff gap 0, diffusion/spectral gap 0.0005, wave gap 0.134 finite-size residue, static POT ruler joins; all vs the square control at $L = 128$). OBS-1 blinds the observer (no graph, coordinates, or dimension target): $d_O = 2.02$ unprompted, $d^* = 2$ train and test, 8% distance match vs the quotient ($2 \times$ closer than microscopic), 6% local charts, $>95\%$ locality, sheet contrast < 0.03 ; expanders $d^* = 3$. $M_O(\text{J2}) \approx \text{J2/sheet}$; a static-channel floor blocks the full three-way cross-probe at $L = 128$. QUOT-0 gives the mechanism (QUOT0-OPERATIONAL): the symmetric sheet sector transports while the antisymmetric sector is exactly dead under $H = -A$ (MALUS-0 M0-NULL); symmetric replay returns the quotient (d drift 0.022), antisymmetric measures

no geometry, a bilayer control stays two worlds (contrast 0.61), staggered perturbation induces only $O(\varepsilon^2)$ remote signal. The DERIVED rung stays open (four design-error sub-bars, autop-sided, never retuned). Reproduce: `pytesttests/test_obs0.pytests/test_obs0r.pytests/test_obs1.pytests/test_malus.pytests/test_quot.py-q`.

Backreaction ontology and accounting. B couples to connectivity three ways: the derived conjugacy, BR-0 energetic selectivity (BR0-D-SELECTIVE*: zero field exactly flat under sampled moves, excitation opens channels — re-read as the no-information limit, not the vacuum), BR-2 quadrature ($B \sim \cos$, $J \sim \sin$), and the BR-2.5 sum map ($\psi_{[uv]} = \psi_u + \psi_v$, $\Delta\|\psi\|^2 = +2B_{uv}$). Admitted primitive: local $u-v \leftrightarrow [uv]$ (BR25-ONTOLOGY); remote relocation demoted to diagnostic; splits many-to-one (powers of two + amplitude degeneracy; loss $|a-b|^2/2$ exact) with selection debt; BR1-FLAT files neutral quiescence. CONS-0 + BR-2.6 close the books conditionally (CONS0-PARTIAL, BR26-ACCOUNTED): triangle-free cycle rank tracks the contraction count, the ledger closes on $B = B_*(c)$ (six families, 10^{-13}), the ΔE formula needs more than B ; no linear conserved combination, no graph reservoir, split conservation never selects. BR-2.7 derives the no-mode verdict (BR27-NO-MODE): downhill scans do not fire, spectral radius exactly 1, no deformation coordinate in a binary graph — one further primitive postulate required; strong stop on rate shopping. Reproduce:

```
pytesttests/test_backreaction.pytests/test_phase.py-q
pytesttests/test_rigidity.pytests/test_contraction.py-q
pytesttests/test_conservation.pytests/test_accounting.py-q
pytesttests/test_stability.py-q
```

Dynamics completion (debt-filing). U0: three contraction-only laws pass the structural gates but splits are unrealized and ties are theorematic at zero field (U0-INCOMPLETE). TIME-0: exact census (143 classes, 102245 pairs) gives unique-history 0.051 below the 0.2 bar with the $T=2$ artifact dissolved (TIME0-NUL; reversibility intact). RAND-0: coherent stochastic apparatus, underived measure (RAND0-MEASURE-DEBT: orbit vs micro uniforms disagree, vacuum not quiescent at one-half). MEASURE-0: 544 cells, all four frozen debt-reasons trigger, CLOSED reachable but not reached (MEASURE0-DEBT). Synthesis: admissible histories known, weight $\mu(\Gamma)$ missing. SYM-0 settles the counting list (SYM0-CLOSED, 1973 cells, 8/8 gates): $X_{\text{red}} = X/(\text{relabel} \times U(1))$ with d_{FG} ; no Born rule. Reproduce: `pytesttests/test_u0.pytests/test_time0.pytests/test_rand0.pytests/test_measure0.pytests/test_sym0.py-q`.

Vacuum family. VAC-FIELD-0 (frozen (J2, $H=-A$), spectrum/symmetry-fixed candidates): VPLUS ($E=-8$, flat ledger, P+, propagating, quotient-visible), VPI ($E=+8$, one-sided ledger, P+), VMINUS ($E=0$, symmetric ledger, P-, joint but decoupled) reach the full JOINT ladder; $\psi = 0$ stops at BACKGROUND ($B_{\text{max}} = 0$, phase undefined everywhere) — the no-information limit, not the vacuum (VACFIELD0-JOINT). Perturbations identical on all four (speed 1.9204, bitwise $\delta\psi$). VAC-EXC-0: $\delta\psi(t) = U(t)\delta 0$ exact ($\leq 1.2 \times 10^{-13}$), bitwise-identical across all vacua (VACEXC0-COMplete, 241 tasks). VAC-COMP-0: even L gives two isolated extremal rays plus a hidden real RP^1 JOINT circle (VMINUS-VSTAG span, $B=0$ BACKGROUND points) $\times$ amplitude ray (10^{-3} – 10^3), $\pi_0 = 3$; odd L keeps VPLUS + VMINUS only ($\pi_0 = 2$); generic eigenstates excluded; mixed beats at $dE = 16/8/8$ (VACCOMP0-COMplete). VAC-SELECT-0: predicted refusal — MEASURE gate unready, 0/23 headline stages ran (VACSEL0-NOMEASURE). Reproduce:

```
pytesttests/test_vacfield.pytests/test_vacexc.py-q
pytesttests/test_vaccomp.pytests/test_vacselect.py-q
```

Hidden sector, linearity, response, zeros, gravity null. HIDDEN-0: hidden transformations distinguish locally (5–6 orders above bar) with remotes at $\leq 5.2 \times 10^{-15}$ and POT remote

exactly 0.0 (HIDDEN0-SEPARATED, 279/279). HIDDEN-BR: equal-energy, wave+POT-blind matched states carry different B (different dE/dA), with 1346 strictly opposite-sign virtual contraction ledgers; a zero-energy pure-hidden state carries a nontrivial ledger — all virtual, no graph operation executed (HBR0-SIGNREV, 214/214). FIELD-0: 57 collision cells hold $I = 0$ exactly (FIELD0-LINEAR) through order-50–80 false accelerations and a forceful-looking static arrangement (FIELD0-APPARENT); a future force must show $I > 0$. RESPONSE-0: field front $v = 7.95$ ($v/8 = 0.993$), quadratic fronts $v \approx 5.94$, laws $|\delta\psi| \sim r^{-0.50}$, $\delta\rho \sim r^{-1.00}$, $\delta J \sim r^{-0.95}$, bipartite B -blindness pinned with the two-sublattice sharpening (RESPONSE0-KERNEL-BANKED, 45 cells). BG-RESP-0: $\chi_+ \neq \chi_\pi \neq \chi_-$ ($\sqrt{88}$ pairwise, rank $2N-1$, null exactly global phase), $\chi_{\text{ZERO}} = 0$ (BGRESP0-COMPLETE, 77 tasks; susceptibility is math response only). ZERO-0: exact zeros are interference-nodal (1588 events at phase π) or eigen-nodal (only persistent class); generic states give zero certified zeros in 1900+ rows; winding slips via bond phase-slip at $|\Delta\theta| \approx \pi$. GRAV-0: no tested strictly local graph-only update both preserves J2 and carries a far disturbance — the carrier must come from coupled (G, ψ) dynamics; the L1 weak-field interface is not retracted. Reproduce:

```
pytesttests/test_hidden.pytests/test_hiddenbr.py-q
pytesttests/test_field0.pytests/test_response.py-q
pytesttests/test_bgresp.pytests/test_zero.py-q
pytesttests/test_zero0_campaign.pytests/test_grav0.py-q
```

VAC-0 completion (substrate classes). The held-out H/I rungs close the campaign as VAC-0 FINAL MIXED: per phenomenon, generic laws (A, D, H-core existence 16/18 on 27/27, I finite-range label 26/27, J algebra 27/27 exact, MZ at LAW level) or class members (F square class, E 2D-ordered class, H turn-on 15/27 broad class, G square-grade class, J-useful 13/27 class), with I ξ scale family-clustered (CLASS-structured). Nothing requires uniquely J2; degree-preserving rewiring destroys F/H/G/DE-validity (degree-only vacuum excluded) while alternatives survive everywhere (square/quotient, +tri/rr3, +hex/ring). Substrate uniqueness dissolves into per-phenomenon classes. Reproduce: `pytesttests/test_vac0_battery.py-q`.

Split anatomy and rewire null. SPLIT-0 decomposes every tested inverse into forced-plus-residual ($M + \xi \leftrightarrow X$, 824/824 roundtrips, minimality 76/76 with two-way ablation witnesses): the covariant residual $\xi = (\text{cover}, d)$ is exactly what the merged state omits. The full inverse is never a singleton (continuous fiber, d_{cont} 1–2); the halves-restricted inverse is deterministic exactly on the 4 isolated-node cells; reverse support holds exactly on the halves subset (36 rev / 24 graph-only / 0 one-way), correcting 3 cells against the MEASURE-0A signature proxy (J-multiset edge-orientation artifact, pinned). No measure derived (SPLIT0-MIXED). REWIRE-0 closes the selector question for the local degree-preserving move (REWIRE0-DEGENERATE, 24/24 J2-L4 states DEGENERATE-or-ABSENT under all six exact principles, scale-persistent L4 $\rightarrow$ L28 with n_{phys} 9792/11291/6756; exact-energy selection additionally ulp-fragile, 1389 violations filed as representation-robustness null): no earned local, covariant, zero-parameter rule selects a rewire on any nontrivial state. Reproduce:

```
pytesttests/test_split0.py-q
pytesttests/test_rewire0.py-q
```

Merge update, reservoir account, trigger census. MERGE-0 characterises selected-edge contraction as a unique covariant deterministic primitive ($R \times U(1)$ exact on all 325 events incl. 87 annihilations, 8 substrates) with exact ledger ($dQ = 2B$, $dE = P1 + P2$ with $P3 = P4 = 0$, support $2 + n_{\text{cross}}$) — while no existing variable closes the energy reservoir on any tested class (0/80 coefficient tuples; MERGE0-DETERMINISTIC + MERGE0-ACCOUNT-DEBT, 38/38). HIDDEN-BR SIGNREV survives execution (2/24 on-support opposite-sign executed ledgers); VPI all- $B < 0$ yet

all- $dE < 0$; favourable orderings on 19/30 scans, still no firing rule. RESERVOIR-0 identifies the missing account $R_{\text{merge}} = -(\Delta E_\psi + \Delta E_G)$ as an exact nontrivial zero-parameter function of the lost inverse information $R(d) = A + |d|^2/2 + \text{Re}(\bar{d}W)$ to $\leq 10^{-9}$ on 26992 fiber rows, both ξ legs varying, swap bitwise, $R_{\text{split}} = -R_{\text{merge}}$, one-neighbourhood-local, disjoint-additive (8.9×10^{-16}), confluent (RES0-XI, 68/68) — and R never vanishes on the halves subset (964 rows, min 0.167) though no field information is lost there: energy debt $\neq$ information debt. TRIGGER-0 censuses the trigger side under strict no-shopping (TRIGGER0-CONDITION, 75/77): 19 of 21 earned exact local predicates survive as CANDIDATE-CONDITION on 257668 edges (BRIDGE NONLOCAL, BAL_R2 VACUOUS at 0/257668; BAL_R1 on exactly 6 hidden-dipole bonds; BJ.ZERO $\iff$ ZERO_MIN exact), with zero dynamical implications in the structural audit. No TRIGGER-1 shopping follows (prereg binding). Reproduce:

```
pytesttests/test_merge0.py-q
pytesttests/test_reservoir0.py-q
pytesttests/test_trigger0.py-q
```

Fiber measure debt and probability-free books. FIBER-0 asks whether earned physics constrains a normalized measure on the SPLIT-0 inverse fiber (FIBER0-DEBT, 22/22): two inequivalent normalized closed-form rivals satisfy every earned constraint (quotient, swap, transport, locality, collapse, ledgers, energy, vacuum, hidden, factorization, scheduler, history), differing with $TV > 0$ on 79 cells; DERIVED was reachable (residual 0) and did not occur (residual 520 = 206 + 168 inter-orbit + 83 radial + 63 angular). Cover measure unique only on the $d = 0$ cells (4/76); factorization a choice (correlated rival TV 0.25); TIME0-NULL survives fiber weights; zero fitted params (AST firewall clean). Six sub-debts filed by name. INFO-0 closes the probability-free books exactly (INFO0-MATCHED, 19/19 HARD, 259 cells, 0 failures): canonical pred == succ 143/143 (mirror theorem, no SEPARATED), timed-skeleton identity, schedulers all $m!$ exact, hidden books green, banked TIME-0 recomputed exactly; the 16/44 labeled gauge audit is descriptive (labels are redundancy; the quotient restores symmetry). No $-\sum p \log p$ anywhere; the weight is still missing. Reproduce:

```
pytesttests/test_fiber0.py-q
pytesttests/test_info0.py-q
```

Reversible store. STORE-0 asks whether the merge-discarded information can be kept in a minimal local hidden store (STORE0-REVERSIBLE, 46/46): the full $\xi = (\text{cover}, d)$ store recovers the exact physical predecessor (5784/5784 fiber rows valid roundtrips over 235 forward events on 79 cells) and closes the merge/split energy account from the same stored content (closure errors $< 3 \times 10^{-14}$, RES0 formula transcribed and cross-checked bitwise), with 5 stored sequences reversed exactly, disjoint factorization with additive R , and hidden/vacuum/texture/excitation batteries green. Minimality is earned: cover-only fails everywhere, d -only fails on all multi-class cells, scalar- R fails on all 78 non-injective cells (59 circle + 19 cross analytic witnesses; 1 proven-injective cell) — ontology: discrete c plus d required. No split-fiber sampling is needed given the complete microscopic state (FIBER-0 rival weightings unneeded on store paths); the store is kept information, not derived selection, and nothing writes, prices, or erases the record. Reproduce: `pytesttests/test_store0.py-q`.

Operational 3D vacuum. DIM-3-0 lifts J2 to J3 ($J3 = Z^3 \rtimes Z2$, $x \leftrightarrow y$, 12 gens) with $H = -A$ unchanged (DIM3-GEOMETRIC, ladder-literal NOT3D via an invalidated trigger): the lift is exactly quotient-cubic (mult 4, $[H, S] = 0$, dead antisymmetric sector, Bloch $\epsilon = -4 \sum \cos$ with isotropic Γ Hessian), the metric reveal confirms 3D quotient geometry (DIST match 0.04–0.07 vs 0.30 bar, quotient-vs-micro factor 2.1–2.6 at L12+, MDS-3 charts 0.09–0.22 pass while MDS-2

0.35–0.43 fail exactly as preregistered, sheet contrast 0.0), and far-field spreading reads r^{-1}/r^{-2} at L20+ with J3 identical to the cubic control on every readout. But blind dimension reads $d^* = 2$ on known-3D data three mechanism-proven ways: d^* structurally capped at 2 (majority-distortion rule stops at MDS-2), arrival-time supralinearity ($M \sim r^{1.48}$ on J3 and cubic alike, predicting the R^2 volume growth to 1%), and static-range P -compression ($\propto 1/\xi$); fronts ride at 2/3 of bound on J3 and cubic alike (forerunner artifact; packets at Bloch speed). No post-data bar moved; 3D-calibrated rulers are DIM-3-1 work. Reproduce: `pytesttests/test_dim3.py-q`.

Vacuum domains, textures, stability. VAC-DOMAIN-0: every fixed-geometry interface between disconnected JOINT components (VPLUS/VPI, VPLUS/hidden $\times 4$, VPI/hidden $\times 4$, both orientations, $L = 28$) emits ballistic $|\delta\rho|$ fronts ($v \approx 5\text{--}6.1$, R^2 0.89–0.92, reach 4–6 cells) while the S-step persists (0.999) and both bulk plateaus hold (drift 0.0); hidden-hidden joins exactly stationary (D-FLAT); P_- residue frozen at weight 1/2; spectral superposition exact with witness $I = 0$ (VACDOMAIN-RADIATIVE, 7/7, 204 tasks). No wall particle or vacuum preference (firewall held). VAC-TEXTURE-0: hidden JOINT orientation can vary spatially staying exactly vacuum-like ($P_- E_0$, $w_{\text{sym}} = 0.0$, $\|H\psi\| = 0.0$, $E = 0.0$ bitwise, no P_+ , frozen) while gradients stay relationally real (local $D = 6.4 \times 10^{-4}$, coarse 1.3×10^{-3} , ledger separation, B -scaling slope +0.173, $Q = a^2$ exact) and dynamically void (symmetric amplitude 0.0, FIELD-0 witness 1.6×10^{-17} , packets exact-split and ballistic at $v = 1.92$; VACTEXTURE-GRADIENT, 14/14, 28 specs). VAC-STAB-0: small protected perturbations stay small on all six backgrounds to $T = 4000$ (max C-ratio 3.5 vs trigger 50, propagating-seed 0.499 below own initial 0.704, positive protection margins, 303 cross-background wraps, frac-sup-B slope 2.0000 exact over $10^{-3}\text{--}10^3$; VACSTAB0-ROBUST, 9/9, 180 tasks, 0 frag cells). No structural claim, none ranked. Reproduce:

```

pytesttests/test_vacdomain.py-q
pytesttests/test_vactexture.py-q
pytesttests/test_vacstab.py-q

```

Sources as boundary data. SOURCE-0: POT’s stationary field IS the driven counterpart of the RESPONSE kernel (K1 dev 0.024–0.037 within the 0.1 bar, K2 exact to 6.6×10^{-12}). A persistent source is boundary data $s(t)$ on $\delta\psi$ in the vacuum frame; the carrier is vacuum-independent with per-vacuum response via banked χ (42/42). Switch-ON always radiates ($v = 5.5\text{--}5.6$); the falsified blanket prediction was release fronts on AMP legs — AMP/VPLUS release is silent by theorem (steady state $c \cdot u_+$ to 7.4×10^{-15}) and AMP/VMINUS release is beating-dominated and frontless (SOURCE0-INCOMPLETE, 9/10, 74/74 records; INCOMPLETE = falsified frozen prediction with complete data). Rule: switch-OFF radiates only on mismatch with free evolution. $R_G \rightarrow dG$ still blocked by MEASURE0-DEBT. Reproduce: `pytesttests/test_source0.py-q`.

Substrate structure and scale. SUBSTRATE-CLASS-0 characterises the VAC-0 class on a frozen 31-cell battery (SUBCLASS0-PARTIAL, 31/31 certificates): E, H-shell, and G admit exact minimal rules each in BOTH spectral and simple-combinatorial form ($E = \{\text{disp_dim_2}\}$ or $\{\text{superlinear_ball}\}$; H-shell = $\{\text{not_hex}\}$ or $\{\text{not_hex_local}\}$; $G = \{\text{square_class}\}$, the frozen plaquette form non-minimal), while F, H-TAU, and J-useful provably admit no exact rule in frozen space (F: the two J2-swap seeds share identical feature vectors with split labels; H-TAU: the ring floor artifact plus the rr3 puzzle; best rule diameter, 4 errors). Spectral vs combinatorial ties on every exact component, so SUBCLASS0-SPECTRAL is refused; no unified cross-component condition exists; sheets are necessary for nothing tested; rewires move 19–25 descriptors at fixed degree. One flag stands: VAC-0 prose says J-useful 13/27, the JSON record counts 12 (JSON authoritative, prose for erratum). SCALE-0 banks the large- L asymptotics (SCALE0-BANKED, 178/178, $L = 64\text{--}512$): d_H 1.9098 $\rightarrow$ 1.9725 monotone toward 2 with j2 = sq to machine precision, d_s L -independent, exact forms upheld (QUOT anti/sheet remote exactly 0.0 to L512, VACEXC deviations exactly 0.0),

velocities exactly L -independent, POT ξ /range exact at all L , packet speed vacuum-independent (1.9028 in all 4 vacua). Reproduce:

```
pytesttests/test_subclass0.py-q
pytesttests/test_scale0.py-q
```

Frozen store, event-local readout. Q-DYN-0 tests $\dot{Q} = 0$ under fixed-graph waiting (QDYN0-INCOMPLETE, 33/35, autopsy-resolved): Q bitwise identical on all 414 rungs with current-M reversal and the semigroup exact, while the two red legs are premise errors — merged states are not eigenstates of the post-merge $H(G_2)$ (residuals 0.094–0.458), and the $\text{Re}(\bar{d}W)$ read-out provably rotates under phase flow for $d \neq 0$ cells (true-eigenvector spread 1.76 vs 3×10^{-13} at $d = 0$). Q-DYN-0b re-tests with corrected legs (QDYN0B-EVENT-LOCAL, 51/51): Q frozen on all 414 rungs (U(1) law to 4.4×10^{-16}), the RES0-XI functional R earned as an event-local accounting functional (E_{aug} drifts to 2.05 while $E_\psi + E_G$ holds to 2.4×10^{-12} ; inverse-split account equals the negative of the CURRENT account to 3.1×10^{-15} ; closed cycles return to 5.4×10^{-15}). Zero evidence for Q dynamics; no further Q-dynamics campaign justified; event timing stays a separate primitive debt. Reproduce:

```
pytesttests/test_qdyn0.py-q
pytesttests/test_qdyn0b.py-q
```

Collapsed-region census, no entropy law. BH-ENT-0 censuses exact joint graph-field-store orbits on collapsed regions (BHENT0-UNCLASSIFIED, 153/153): 8/67/701/10047/218083 on paths P2–P6 at fixed boundary ($\log_2$ slope 3.67, $R^2 = 0.9945$ — yet strict VOLUME rejected by second differences plus a quadratic F-test at $p = 0.0052$, BOUNDARY rejected by growth at fixed boundary, MIXED rejected with $R^2 = 0.928$ off-band). Interior-graph multiplicity dominates and is volume-superlinear ($\sim 2^{n^2/2}/n!$ quotient-corrected); boundary wiring saturates (path leg wirings = 2). Matched sign pairs are 36/36 near+far blind on J2 (VPLUS and VPI, L6 and L28) vs 0/12 paths and 0/16 square: hidden-sector field multiplicity is exterior-blind and extensive in $|R|$, while graph multiplicity is dynamically exterior-visible (POT distinguishes graph pairs). No asymptotic law supported, no S_{BH} identification (firewall held), no finite fiber count (MEASURE0-DEBT). Reproduce: `pytesttests/test_bhent.py-q`.

Repaired 3D rulers. DIM-3-1 repairs the DIM-3-0 estimators and measures J3 blind (DIM31-GEOMETRIC): dimension reads 3 through six independent channels (arrival-time transfer + direct medians 3.458/3.552/3.757 at L16/20/24, d^* 9/9, charts 9/9, static L20 2.932 / L24 3.135 with J3/cubic static identity to six decimals, 3D spread laws, packets at Bloch speed) — but the cubic-gamma transfer the arrival channel relies on disagrees with J3’s own gamma beyond the calibrated bar at L16 (by 0.007) and L20 (by 0.042), passing barely at L24. By the frozen all-gates rule the verdict is GEOMETRIC, not OPERATIONAL: every dimension-reading channel is unanimous at 3; what failed is a transfer validation leg, decaying with L . Reproduce: `pytesttests/test_dim31.py-q`.

Random weave: no 3D core. WEAVE-0 tests Poisson-sheet weaves over a coupling ladder for a joint volume+spectral 3D core (WEAVE0-INCOMPLETE, ambiguous: no core, not stably 2D/nongeometric; 270/270 grid cells): the apparatus validates (B/C controls read C0 as 2D, C4/C3 as 3D; C5 never matches, 0/8), but volume reads 3D-side only at $\lambda = 0.01$ (B-head 5/8; 2/8 at 0.005, 0/8 above with overshoot past 3.5) while spectral never reads 3D jointly (0/8 every rung; 3D only at $\lambda = 0.16$ where volume explodes) — no B+C joint cell, `core_lams` empty. E/F and H legs filed apparatus-invalid, not repaired. Randomness alone, as woven here, does not make 3D. Reproduce: `pytesttests/test_weave0.py-q`.

Store/qubit identity and blind-store dimension. Q-INFO-0 asks whether the STORE-derived $h_2(P_-)$ is algebraically identical to the banked qubit functional under the exact Hadamard map (QINFO0-IDENTICAL, 16/17 — the red gate is the equiv-rule, red by design): $\max |h_2 - S_{\text{banked}}| = 1.1 \times 10^{-16}$ (machine precision vs a 10^{-9} bar), same base 2, same $0 \log 0 = 0$ convention; multi-entry H_Q filed separately per the no-relations firewall; the factor-two audit reads (d_R, d_I) as one complex amplitude. Handoff: reuse the earned qubit measure for isolated Q modes; no new entropy. BH-Q-ENT-0 then censuses the exterior-blind continuous STORE dimension (BHQENT0-UNCLASSIFIED, 34/34, the pre-data prediction): joint D is 0 on paths/squares, $2n-4$ on stars, n on J2 disks (topology, not (n, b) , controls blindness); all law gates miss (boundary $R^2 = 0.68$, volume 0.89, $b \log b$ 0.72), reported unmoved; blind weight boundary-adjacent; complex- d factor two resolved; STORE leg POT-blind where graph legs are visible. No measure, no bits, no entropy. Reproduce:

```
pytesttests/test_qinfo0.py-q
pytesttests/test_bhqent0.py-q
```

First area law: boundary Q-information at maximum density. BH-Q-AREA-0 embeds a complete-graph core in a J3-ball ladder ($r = 1..10$) and censuses banked QINFO0 s_Q on 132–5304 boundary channels per rung (BHQAREA0-MAX, 22/22): $S_Q^\partial = \kappa^* A + o(A)$ with $\kappa^* = 4.2207$ bits/unit-area, $\sigma^* = 4.2208$ channels/unit-area, $h^* = 0.999990$ bits/channel ($\bar{h}$ ladder monotone $0.72 \rightarrow 1$; power $p(S \text{ vs } A) = 0.92$; volume-density strictly decreasing). Doubly selected: patterns give $S = 0$ exactly (vacuum-on-BH-geometry property), and the plain-J3 control gives a non-MAX law ($\kappa = 0.43$, $h \approx 0.10$) with disjoint distributions ($KS = 1.0$) — the complete core selects MAX. One disclosed amendment (domain-vs-domain G–J; v1 NONUNIVERSAL preserved, freeze byte-identical); conditional $a/\ell_P = 3.42$ with NO agreement claim. Firewall holds: isolated boundary information only — not Bekenstein–Hawking entropy, not a horizon, not Hawking radiation. Reproduce: `pytesttests/test_bhqarea0.py-q`.

Jets closed, flow never forces. JET-1 repairs JET-0’s malformed aggregate `compat_keys` comparison with the correct per-rung bitwise check (JET1-NULL, 13/13): 0 mismatches over 1152 bits, searched/ever/orbit/search-count agreement on all 15 cells, banked JET0-INCOMPLETE 31/32 reproduced exactly, $n_{\text{genuine}} = 0$ with 0 crossings on 18 ordinary trajectories under unchanged definitions — no surface promoted; filed stop rule (no JET-2). EVENT-0 censuses structural event necessity (EVENT0-EQUIV, 33/33): all 426 rungs of all 71 fixed- G trajectories stay valid ($t^* = +\infty$), Q bitwise frozen, reversal exact — but 61 exact nontrivial same- N rewire-equivalence orbits exist on small symmetric graphs (J2-L4 ruled out graph-first, 0/9792 cospectral neighbors), with zero firing implications and 2493 crossing edges filed as TRIGGER-0-consistent conditions. EQUIV, not FORCED. Reproduce:

```
pytesttests/test_jet0.pytests/test_jet1.py-q
pytesttests/test_event0.py-q
```

Debts (by name; D14/D15 taken). Substrate classes (VAC-0 Final MIXED: F mechanism, VAC-0Q held-out battery, unified H-turnon property; SUBCLASS0-PARTIAL: no exact F/H-TAU/J rule in frozen space; WEAVE0-INCOMPLETE: no 3D core on random weaves); 3D transfer validation (DIM-3-1: dimension 3 on all channels, cubic-gamma transfer fails `tol_agree` at L16/L20); vacuum-field member (needs the history measure first); history measure $\mu(\Gamma)$ (reversible, both readings; fiber rivals differ on 79 cells); structural kinetics (firing postulate; 19 conditions, 0 implications; EVENT0 flow never forces, 61 orbits, 0 implications; JET1-NULL stop rule); split information (fiber measure, 520 filed dof; kept- ξ store filed as kept, not derived; Q-DYN-0b: Q frozen, R event-local, timing stays debt; QINFO0: h_2 is the banked qubit fn to $1.1\text{e-}16$); BH entropy

(BHQAREA0-MAX isolated-info area law; thermodynamics open: temperature, first law, Hawking flux); neutral quiescence; matter (beat $I = 0$); gravity carrier (coupled dynamics). Blocked until then: complete graph dynamics, formation under it, matter, the carrier, decay, and force phenomenology. Full ladder: `docs/scaffolding-history.md`; model statement: `docs/model.md` §10.

Table 1: Every number with a status: input = assumed, fit = calibrated, measured = code output, derived = follows from premises.

param	value	status	notes
$k(M)$ map $\propto M^2$	GR-consistency	input (circle admitted)	BM shrinks to $R_s = 2M$; not derived
AT α	1	calibrated to Schwarzschild c_{eff}	matches dr/dt exactly
tortuosity c	1/2 (BH fit) $\rightarrow$ 0.44–0.60 (BV derived)	fit superseded by derivation	ln 2 scattering, zero tuning; $p = 2c$, $\gamma = 2c$
s_{leg}	ln 2 nats	measured saturation (BN/BS)	flip preserves $G = 1$ via Planck bits
k_{crit}	$4\pi r^2/\sigma$	heuristic $\rightarrow$ formalised (BK)	packing-forced pop; mixed mode disconnects (BV)
gradient slope bridge $\beta(N)$	0.015 1.5@300; 1.28@600; 1.24@1020; 0.99@4k; 0.87@8k; 0.74@16k	fitted, labelled recalibrated per N	$p_{\text{adj}}(s) = 0.85 + 0.015s$ log-linear over 6 points; derivation queued
$w = 1.953$	solved from cancellation	fitted (solved, not tuned)	$c_{\text{tot}} = c_1 + wc_2 = 4.8695$ at $p = 0.92$
shed $e + \epsilon$	$0.5 \rightarrow 0.416 + 10\%$	calibrated once on AT2017gfo	M_{ej} norm-independent
shed $\epsilon(M, a)$	0.0168 universal (Eq. 2 main)	calibrated + extrapolated (D8)	mass/ratio independence underived; BBH extension tested, not fitted
upper-gap null	no feature at $44 M_\odot$	negative prediction (measured smooth)	$k/\text{spin}/\text{Love}/\text{congestion}$ all smooth; edge left to stellar physics
GW190814	$P(\text{detect}) \approx 0.68\text{--}0.88$	tension, POSSIS queued (D7)	non-detection at $p \sim 0.12\text{--}0.32$; epoch audit in massgaps
vacuum P0'	isostatic 2D, $\langle z \rangle = 4$	postulated (v0.6)	BH = max-tension extreme; $d_I = 2$ predicted; M_O reconstruction
d_{eff} protocol	$d \ln V / d \ln r$ bracket	measured (25 tests)	shortest below + diffusion above on lattices; shells fail (S11)
D10 tension map	GR dip/overshoot/ $\rightarrow 3^+$	GR computed; graph open	bare shortest-path rejected (inverted); D11 tail tension-dep. ($\chi_1 1/r^2$, $\chi_2 1/r$, χ_5 crossover $q \rightarrow 1.1$); κ -profile closed (disk, mild-disk confirm)
D12 reconstruction	$\forall M_O \in \mathcal{A}_{\text{macro}}$ same IR	open program	P4 audit 1,2,3,5 pass, 4 partial; vantage axis filed; static depth neg. (vol+MDS); MDS bar per-substrate (del bowl)
D13 causal order	D13.0 control measured; stages 1–4 open	static control measured (S11)	V_G , $8r+4$ law; family (tri/hex/Del/Lloyd/Gab/kNN/medial +, gated-wall/rewired -); sweep: $N^* O(1)$ range, $f^* \rightarrow 0$ hypoth.; spectral d_s band; twins C3
U -rule search	locality protection by U	third pass: blind- U + substrate census	blind- U + admissibility; 7-substrate coord. verdicts; self-cal. census
D14 pricing origin	hierarchy gain ~ 350	debt by elimination (measured)	basin + feedback OK; formation: DARK, NO-coex (N^*), SSB-1, St-0 NULL
J2 micro/macro probe	quotient = square lattice	measured; working vacuum (v5.4)	quotient $4r$ exact; bipartite; perturb. family-typical; orient. absent (32 runs)

Table 2: Honesty ledger, continued: relational field program (S12). Same status words.

param	value	status	notes
field postulates	$\psi=r+is$ per node, $H=-A$	postulated FP1 / working law FP2	one-way $G \rightarrow \psi$ map; not derived (S12)
field anatomy	B/J conjugacy	derived LAW (all simple graphs)	VAC-0A A1–A6 on hostile graphs; quadrature $B \sim \cos$, $J/2 \sim \sin$
substrate class	$\partial E/\partial A = -2B$ per-phenomenon LAW/CLASS split	measured (VAC-0 Final MIXED)	nothing requires uniquely J2; degree-only excluded; exact E/H-shell/G rules, F/H-TAU impossible in frozen space (SUBCLASS0-PARTIAL); large- L banked, j2 = sq (SCALE0); J3 quotient-cubic, repaired rulers read 3, transfer open (DIM31-GEOMETRIC); no 3D core on random weaves (WEAVE0-INCOMPLETE) (S12)
EM reading	ψ is ordinary electromag- netism	rejected (EM1- FALSIFIED)	range saturates $\xi \approx 0.53$; 1 scalar mode; no gauge redundancy; drift $v \cdot q$
geometry change	contraction/split $u-$ $v \leftrightarrow [uv]$	ontology admitted; firing absent	ledger exact; BR27-NO-MODE; SPLIT0 ξ residual; MERGE0 update unique; RES0 $R=f(\xi)$; STORE0 kept- ξ store reversible; Q-DYN-0b Q frozen, R event-local; QINFO0 $h2$ = banked qubit fn; TRIGGER0 19 cond. 0 imply; REWIRE0 no selector; JET1-NULL closes jets; EVENT0 flow never forces (S12)
dynamics completion	U_0 /TIME-0/RAND- 0/MEASURE-0/FIBER- 0/INFO-0	all file the measure debt; books exact	no unique $\mu(\Gamma)$ forced; SYM-0 counting X_{red} ; BHENT0 exact orbits, no asymptotic law; BHQENT0 topology- D , no law; BHQAREA0 $S_Q = \kappa * A$ MAX (isolated info only); fiber rivals differ, 520 dof; INFO0 pred==succ 143/143 (S12)
observer quotient	$M_O(J2) \approx J2/\text{sheet}$, $d_O = 2.02$	measured blind; mechanism opera- tional	6 rulers meet; symmetric sector transports; derived rung open (S12)
vacuum family	VPLUS/VPI/VMINUS joint; $\psi=0$ no-info limit	measured family; selection refused	carrier background-free; manifold censused; VACSEL0-NOMEASURE; domains RADIATIVE; textures GRADIENT; stab. to $T=4000$ (S12)
hidden sector	dead sector stores local info, reverses response	measured (virtual ledgers)	HIDDEN0 separated; HBR0 1346 sign-reversed edges, all virtual (S12)
linearity/response	$I=0$ null; kernel + per- vacuum χ banked	measured (57 + 45 + 77 cells)	fronts 7.95/5.94; $\chi_+ \neq \chi_\pi \neq \chi_-$, $\chi_{\text{ZERO}}=0$ (S12)
sources	persistent $s(t)$ = bound- ary data on $\delta\psi$	measured; blanket release falsified	K1 dev 0.024–0.037; switch-OFF radiates only on mismatch (S12)
$\kappa \rightarrow c_2$ map	$c_2 = p(2p-1)$	ansatz (open deriva- tion)	power-law vs $1/r^2$ disagree cross-applied
evap. isometry V_k	qubit V_t done (D2); graph small- N ED done	large- N open (S3)	$V_t^\dagger V_t = I$ tested; graphvk all:all < 0.25 bits vs chain > 0.4 at $N=8$ (D1)
Kerr M_2 , $g_{t\phi}$, ISCO, QNM	area only	not derived (S3)	$A(M, a)$ assumed; quadrupole/QNM open; GW250114 QNM + GW241011 δ_Q future wires
echo R	10^{-80} ampl., 10^{-160} en- ergy	dimensional esti- mate	$\propto \omega^2$ lattice scattering, not leg S-matrix; ~ 158 orders margin

Table 3: Gradient-shell p measurements: β recalibrated per N (log-linear over 6 points). All 80-graph production rows use exact distances; errors are per-graph std with SEM quoted in text.

N	shells $\times$ per-shell	β	measured p	backend
300	10×30	1.50	0.94 ± 0.06 (4 graphs)	NetworkX exact
600	10×60	1.28	0.96 ± 0.06	NetworkX exact
1020	10×102	1.24	0.913 ± 0.049 , SEM 0.0055, stacked 0.911 (80 graphs)	Floyd+LP exact
4000	CSR-direct	0.99	0.9315 ± 0.0032 (80 graphs)	CPU + Sinkhorn OR
8000	CSR-direct	0.87	0.9382 ± 0.0030 (80 graphs)	pod + Sinkhorn OR
16000	CSR-direct	0.74	0.9137 ± 0.0022 (80 graphs)	GPU farm + torch OR

Table 4: PPN status: only γ is claimed. β is coordinate-confused and not quoted.

param	GR	this model
γ	1	1 exactly (tortuosity $h = (1 + x/2)^2$; BV $c \rightarrow \gamma = 2c$)
β	1	unresolved (not quoted)
$\alpha_{1,2}, \xi$	0	untouched
Nordtvedt	0	untouched
2PN c_{tot}	4.8695	4.8693 at $p = 0.92$ (J0737 0.1 σ)

Table 5: What exists today: no qualifying test yet. Only multi-detector, $< 100 \text{ deg}^2$ gap events count.

event	masses	distance	localisation / EM	verdict
GW190814	$23.2 + 2.6$ ($q \sim 0.11$)	241^{+41}_{-45} Mpc	23 deg^2 (3-det.); CFHT g $22.8@1.7\text{d}$ 65.5%, GROWTH i 6 epochs to 98%	tension: $P(\text{detect}) \approx 0.68$ – 0.88 , $p \sim 0.12$; not excluded
GW230529	$2.5\text{--}4.5 + 1.2\text{--}2.0$	~ 197 Mpc	$24,200 \text{ deg}^2$ (single-det.); ZTF 7% to $g = 21.1$	not a test: 93% unsearched; $m_g \sim 21.2$ at depth edge
S250206dm	mgBH candidate	—	no counterpart (per notes)	unverified: check GCNs before citing